Optimal Carbon Pricing of Closed-Loop Supply Chains: A Global-Optimisation of a Bi-Level Mixed-Integer Problem

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Abstract

We study closed-loop supply chain (CLSC) network design under a government-imposed carbon tax, formulated as a bilevel mixed-integer program (BL-MIP) that makes the regulator–operator interaction explicit. The regulator (leader) selects a discrete carbon tax, lexicographically minimising the induced supply-chain emissions and then levying the lowest tax that attains that minimum; the supply-chain operator (follower) responds by optimising facility location, capacity, technology selection, transportation, and forward and reverse product flows to minimise operating cost inclusive of tax. We solve the resulting BL-MIP with a *deterministic bounding procedure* (DBP) that compute the regulator's optimum function between two bounds. A linear-programming (LP) relaxation of the follower provides a lower bound, and the follower's exact integer response provides an upper bound. The DBP iteratively tightens these two bounds until their gap is below ϵ. Because the supply chain chooses among finitely many configurations, its emission response is piecewise constant and saturates: beyond the saturation threshold, further taxation abates nothing. A regulator targeting emissions alone is therefore self-defeating — prone to setting arbitrarily high taxes that yield no additional abatement. The saturation threshold is determined by the installed production and remanufacturing capacity of each network. Because that threshold differs across network designs, the tax varies by an order of magnitude across our computational instances, and no single carbon tax fits all.

Keywords: bilevel optimisation; carbon pricing; closed-loop supply chain network design; environmental regulation; global optimisation; lexicographic optimisation; mixed-integer programming.

1 Introduction

Climate change and the regulatory pressure for sustainable industrial practices have prompted governments worldwide to deploy market-based instruments for controlling greenhouse-gas (GHG) emissions. Among these, *carbon taxation* is widely regarded as one of the most transparent and economically efficient policy tools: by attaching a monetary price to each unit of CO_2-equivalent emitted, it internalises the negative externality of pollution and creates incentives for supply chains to redesign networks, adopt cleaner production technologies, and invest in reverse logistics [Nordhaus, 2007, Pigou, 1920]. Evidence from an implemented carbon tax shows that a well-calibrated rate can produce a substantial, empirically verified drop in emissions [Andersson, 2019], although the effectiveness of any given rate is mediated by the structure of the affected supply chains.

Supply chain network design (SCND) is one of the most consequential strategic decisions an organisation makes. Facility location, capacity sizing, technology installation, and transportation contracts typically lock in operating costs and emissions over horizons of several decades [Melo et al., 2009]. When a carbon tax is in force, these long-lived decisions must simultaneously balance cost efficiency, regulatory compliance, and environmental performance. The rise of *closed-loop supply chains* (CLSC), which integrate forward distribution with the recovery, remanufacturing, and reuse of end-of-life items, further expands the design space and

enlarges the opportunities to abate emissions through circular operations [Fleischmann et al., 2001, Govindan et al., 2015].

Crucially, the interaction between the regulator and the supply chain operator is *inherently hierarchical*: the regulator commits first to a tax level, after which the operator optimises its network and operations. Conventional single-level SCND models, including emission-aware variants, take the carbon tax as a fixed parameter and therefore cannot represent the regulator's strategic problem of *choosing* a tax level so as to induce a desirable response from rational, cost-minimising supply chain operators. As a consequence, they may recommend tax rates that are either ineffective (failing to trigger network reconfiguration) or excessive (imposing economic costs out of proportion to the abatement they achieve).

1.1 Motivation

Bilevel optimisation provides a natural framework for modelling Stackelberg leader–follower games [Colson et al., 2007, Dempe, 2002]. In the present setting:

- the **leader** (regulator) selects the carbon tax, with the goal of minimising the supply chain's total GHG emissions and, among all tax levels that achieve that minimum, levying the lowest one;
- the **follower** (supply chain operator) solves a multi-echelon, multi-period CLSC design and operations problem parameterised by the carbon tax, minimising total operating costs inclusive of tax payments.

The resulting BL-MIP is NP-hard even when both levels are linear programs [Jeroslow, 1985]. The presence of binary facility, technology, and capacity decisions in the follower compounds this difficulty and rules out direct application of techniques designed for continuous bilevel programs. Specialised solution methods are therefore required.

1.2 Contributions

This paper makes three interrelated contributions, each positioned against the closest related work in Section 2.4:

1. We formulate a tactical BL-MIP in which a regulator chooses the carbon tax from a finite grid — minimising induced emissions, then the tax itself — anticipating the optimal response of a multi-echelon, multi-period closed-loop supply chain operator whose facility, technology, and capacity decisions remain integer.
2. We develop a DBP that exploits the leader's finite tax grid directly, combining valid lower bounds from a parametric LP relaxation of the follower with valid upper bounds from exact follower responses at sampled tax levels, with proved finite-time convergence to a global ϵ-optimal solution.
3. Runs on generated benchmark instances show that the optimal tax is strongly instance-specific, that the emissions response saturates beyond a threshold — an over-taxation pathology for a purely emissions-minimising regulator — and that cleaner production/remanufacturing technology, not transport-mode choice, is the binding abatement

lever.

1.3 Paper organisation

The remainder of the paper is structured as follows. Section 2 reviews the related literature on bilevel optimisation, sustainable SCND, and carbon pricing in operations management, and positions our contribution. Section 3 describes the physical network and the hierarchical decision structure. Section 4 states the bilevel mixed-integer mathematical formulation. Section 5 presents the DBP and establishes its convergence. Section 6 reports computational experiments. Section 7 discusses the managerial and policy implications, robustness of the findings, and limitations. Section 8 concludes and outlines directions for further research.

2 Literature Review

This section surveys three streams of research that converge in our contribution: (i) bilevel optimisation and its computational treatment, (ii) closed-loop and sustainable supply chain network design, and (iii) carbon pricing in operations management and environmental economics. We close the section by articulating the research gap that motivates the present paper.

2.1 Bilevel optimisation

Bilevel programming, formalised by Bracken and McGill [1973] in the context of Stackelberg games [von Stackelberg, 1934], has matured into a substantial subfield of mathematical programming. Foundational treatments are provided in the monograph of Dempe [2002] and the survey of Colson et al. [2007]. Even in its simplest, fully linear form, bilevel optimisation is NP-hard [Jeroslow, 1985, Hansen et al., 1992], which has motivated a steady stream of methodological advances.

2.1.1 Exact reformulations

The most common exact route replaces the lower-level problem by its KKT optimality conditions and linearises the resulting complementarity constraints [Fortuny-Amat and McCarl, 1981]. While this approach is conceptually simple, the disjunctive structure of complementarity rapidly degrades scalability, particularly when the follower contains binary variables. Tailored branch-and-bound and branch-and-cut methods that exploit the hierarchical structure have been developed by Moore and Bard [1990] and, in the mixed-integer case, by DeNegre and Ralphs [2011].

2.1.2 Bounding and decomposition

Penalty-based bounds for bilevel linear programs were proposed by Edmunds and Bard [1991]. Subsequent work has produced general-purpose exact solvers for mixed- integer bilevel linear programs (MIBLPs): Fischetti et al. [2017] develop a branch-and-cut algorithm based on

intersection and no-good cuts that solves hundreds of previously open benchmark instances; Lozano and Smith [2017] pursue an alternative, value-function-based reformulation that avoids an explicit single-level recast of the follower; and Tahernejad et al. [2020] describe a generalised branch-and-cut framework, implemented in the open-source solver MibS, that unifies several of these cutting strategies. Kleinert et al. [2021] survey this rapidly developing area, highlighting the role of no-good cuts, disjunctive cuts, and intersection cuts in mixed-integer bilevel programming. Our DBP is in a similar spirit — it bounds the leader's optimal value from below and above and tightens both bounds iteratively — but it exploits the specific structure of a *discrete, finite* leader domain (a tax on a grid) coupled with a parametric mixed-integer follower, rather than relying on generic disjunctive cuts over a continuous leader space.

2.1.3 Bilevel models for network design and pricing

Bilevel optimisation has a long history in network design and pricing problems, where a leader sets tolls, tariffs, or capacities and a follower (or population of followers) reacts through flow or allocation decisions; Brotcorne et al. [2001] is a classical example for toll optimisation on a multicommodity transportation network. Closer to our setting, Yue and You [2017] formulate a Stackelberg game between a supply-chain leader and operator as a mixed-integer bilevel program in which the follower retains discrete facility, technology-selection, and shutdown decisions, solved through a reformulation-and-decomposition algorithm. Their framework establishes that a discrete-follower bilevel SCND model is computationally tractable, but it is not posed as an environmental-regulation problem and the leader does not face a finite policy grid; we build on this precedent in Section 2.4.

2.2 Closed-loop supply chain network design

The SCND literature is extensive; comprehensive reviews are given by Melo et al. [2009], Farahani et al. [2014], and, with explicit attention to sustainability, by Eskandarpour et al. [2015] and Govindan et al. [2015].

2.2.1 Closed-loop models

CLSC models jointly optimise forward and reverse logistics, capturing collection, remanufacturing, and reuse alongside production and distribution decisions. Savaskan et al. [2004] provide the canonical analytical treatment of channel choice for collecting used products for remanufacturing; Fleischmann et al. [2001] propose an early integrated network-design formulation. Subsequent work extends these models along several dimensions: to multi-period, multi-product, and multi-modal settings [Pishvaee et al., 2010, Özceylan et al., 2014]; to the joint design of forward and reverse flows under demand uncertainty [Cardoso et al., 2013]; to multi-period capacitated location with delayed demand satisfaction [Correia and Melo, 2014] and p-robust location under disruption [Snyder and Daskin, 2006]; and to a triple-bottom- line treatment of cost, emissions, and social impact solved via hybrid metaheuristics [Devika et al., 2014]. Robust and stochastic extensions addressing demand and return-rate uncertainty are

presented by Pishvaee et al. [2011] and Keyvanshokooh et al. [2016]. Our follower formulation builds on this stream and extends it through intermodal hub routing in the spirit of strategic hub-network design [Yıldız et al., 2021], technology-specific production and remanufacturing options, and explicit treatment of storage-area capacity.

2.2.2 Carbon aware SCND

A growing body of work embeds emission considerations into network design through emission caps, multi-objective formulations trading off cost and emissions, or explicit tax payments. Chaabane et al. [2012] and Fahimnia et al. [2013] analyse the impact of carbon policies on distribution networks; Mota et al. [2018] integrate sustainability indicators into a multi-product CLSC model with technology selection; and Zhen et al. [2019] pose a bi-objective cost-versus-CO_2 closed-loop network design problem under demand uncertainty. In every one of these studies, however, the carbon price or emission cap is a fixed parameter set by the modeller, not a decision optimised by an explicit regulator.

2.3 Carbon pricing in operations management and environmental economics

The choice between price-based instruments (a tax) and quantity-based instruments (a cap or tradable permits) for controlling an externality is a classical question in environmental economics. Weitzman [1974] shows that the relative efficiency of prices versus quantities depends on the curvature of marginal costs and benefits, and Newell and Pizer [2003] extend the comparison to stock externalities such as greenhouse gases under uncertainty, a setting in which price instruments are generally favoured. Aldy et al. [2010] synthesise the resulting policy-design literature, covering instrument choice, stringency, and dynamic efficiency for climate mitigation. These results motivate our leader's choice of a tax *level* on a fixed instrument (rather than a quantity cap) as the policy variable, and they underlie the over-taxation pathology we document in Section 6.2: a price instrument calibrated purely to a quantity (emissions) target is, in Weitzman's terms, mismatched to a follower whose marginal abatement cost is itself discontinuous and eventually near-vertical.

Within operations management, Tang et al. [2015] and Benjaafar et al. [2013] compare carbon tax, cap-and-trade, and cap-and-offset schemes, characterising their respective effects on production, inventory, and supply-chain decisions. Xu et al. [2016] study a two-echelon supply chain operating under a carbon tax, while Ji et al. [2017] examine a single-manufacturer SCND problem under cap-and-trade, and Zakeri et al. [2015] investigate the joint design of forward and reverse flows under emissions regulation. None of these papers, however, treats the tax level itself as a decision made by an optimising regulator who anticipates the supply chain's response.

2.3.1 Bilevel models with environmental regulation

A small but growing literature applies bilevel optimisation to environmental policy design. Zhang et al. [2021] consider a bilevel model for carbon-quota allocation but restrict the follower

to a linear program with no integer variables, foreclosing facility-level decisions. He et al. [2019] develop a bilevel green supply-chain model focused on pricing rather than network design. Cao et al. [2020] address tax differentiation across regions but treat the supply-chain problem as a continuous flow problem.

2.4 Research gap and positioning

Four observations emerge from the surveyed literature. *First*, the vast majority of carbon-aware SCND studies (Section 2.2) model the tax or cap as a fixed parameter and thus cannot inform the regulator's choice of an effective tax level. *Second*, the existing bilevel formulations of environmental regulation [Zhang et al., 2021, He et al., 2019, Cao et al., 2020] simplify the follower problem to a continuous, single-period setting, omitting integer facility, technology, and capacity decisions that are central to long-run emissions; conversely, the bilevel SCND model that *does* retain a discrete follower [Yue and You, 2017] is not posed as an environmental-policy problem and has no regulator objective over emissions. *Third*, environmental economics clarifies *why* the leader's instrument matters [Weitzman, 1974, Newell and Pizer, 2003, Aldy et al., 2010] but stops at the single-firm, reduced-form level and does not model a multi-echelon network operator as the follower. *Fourth*, even when a bilevel mixed-integer model is proposed elsewhere in the optimisation literature, algorithmic treatment is often limited to KKT reformulations or generic disjunctive cuts [Fortuny-Amat and McCarl, 1981, Fischetti et al., 2017, Kleinert et al., 2021] whose scalability or applicability has not been demonstrated for a discrete leader domain coupled with a multi-period closed-loop follower.

To the best of our knowledge, no prior work simultaneously (i) formulates a *bilevel* carbon-tax problem with a *multi-echelon, multi-period closed-loop* SCND follower incorporating integer facility, technology, capacity, and transportation decisions; and (ii) develops a convergent *deterministic bounding procedure* adapted to the discrete leader domain, together with a lexicographic tie-break that exposes — rather than conceals — the over-taxation pathology implied by Weitzman [1974]'s price-versus-quantity argument. The present paper fills this gap.

3 Problem Description

We study the tactical design of a multi-echelon, multi-period CLSC operating under an external carbon-tax policy. The resulting decision problem is hierarchical: a regulator sets the tax rate, and a supply chain operator designs and operates the network under that rate. Section 3.1 formalizes the physical network and admissible flows; Section 3.2 states the Stackelberg interaction that underpins the bilevel model of Section 4.

3.1 Network and flows

The planning horizon comprises T discrete periods. The physical network consists of five node classes: suppliers ($\mathcal{S}$), factories ($\mathcal{F}$), warehouses ($\mathcal{W}$), hub terminals $\mathcal{H} = \mathcal{H}^{\mathrm{a}} \cup \mathcal{H}^{\mathrm{s}}$ (airports and seaports), and markets ($\mathcal{R}$). Suppliers, factories, warehouses, and hub terminals are *openable*.

Opening a factory or warehouse incurs a construction cost and an emission, both proportional to its installed area – there is no fixed cost or emission for opening a factory or warehouse beyond what its installed area requires. Suppliers and hub terminals are not constructed but *selected*: selecting a supplier incurs a fixed cost, reflecting the footprint of qualifying a new sourcing relationship rather than building a facility, but no emission of its own – a supplier's emission footprint is captured entirely by the raw-material acquisition emission charged on what is actually shipped from it – whereas selecting a hub incurs a fixed access cost but no construction emission, since hub terminals are modeled as outsourced terminal access points. Markets face demand given as a model parameter and act as collection points for end-of-use items.

Three item families circulate in the network. Raw materials ($\mathcal{U}$) are procured from suppliers and consumed at factories. Finished products ($\mathcal{P}$) are manufactured or remanufactured at factories and delivered to markets through warehouses, hubs, or direct links. Recoverable items ($\mathcal{K}$) are collected at markets and returned upstream for remanufacturing, closing the material loop. Forward and reverse flows share the same transportation infrastructure.

Transportation combines a heterogeneous modal portfolio. Road movement uses company-owned truck alternatives, so the model can choose among truck types with different capacities, fixed costs, trip costs, and emission intensities. Long-haul inter-hub transfers use outsourced airplane and boat alternatives, represented by carrier contracts with per-period capacity, variable kg–km costs, and terminal handling costs. Hub terminals operate as intermodal cross-docking facilities: ground inflows are matched by air or sea outflows of the same tonnage, and vice versa.

Production and remanufacturing at each factory rely on a catalogue of technologies ($\mathcal{G}$ and $\mathcal{L}$, respectively), each characterized by unit cost, throughput bounds, emission factor, and a set of compatible products. Finished-product inventories are held at factories and warehouses subject to opening-conditional capacity limits; markets satisfy demand exactly with no carry-over. Returned items enter the reverse channel with a product-specific lag and must be absorbed by remanufacturing at a factory.

Figure 1 depicts the admissible arc families. Solid arcs denote forward flows (direct and warehouse-mediated), dashed arcs denote reverse flows, and grey arcs denote intermodal routes through the hub layer. The two hub nodes in the figure represent the same set $\mathcal{H}$ in its origin and destination roles along an intermodal path. Demand and return-rate bounds are deterministic and known at planning time, consistent with the tactical scope of the problem.

3.2 Bilevel structure

Network design and operation are governed by a Stackelberg game between a regulator and a supply chain operator: the regulator commits to a carbon tax before the operator observes it and responds.

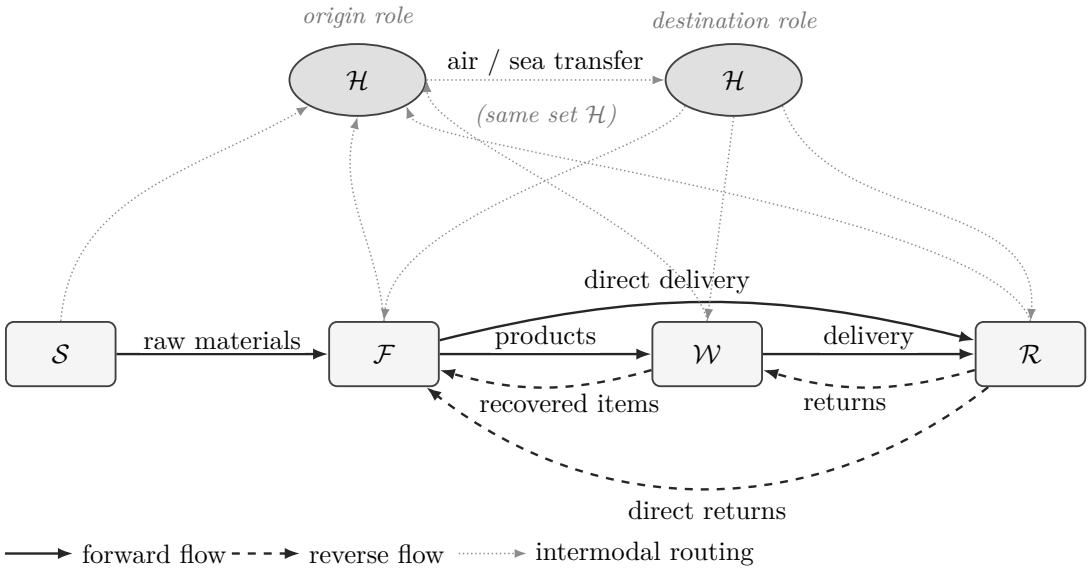


Figure 1. Forward (solid), reverse (dashed), and intermodal (dotted) flows in the CLSC network, connecting suppliers ($\mathcal{S}$), factories ($\mathcal{F}$), warehouses ($\mathcal{W}$), and markets ($\mathcal{R}$). Intermodal routes pass through the hub layer $\mathcal{H}$, shown twice to distinguish its origin and destination roles.

3.2.1 Leader (regulator)

The regulator chooses the carbon tax from a finite, discrete grid of rates, which enables the exact decomposition of Section 5. The regulator's objective is *lexicographic*: first minimize the supply chain's total emissions, and, among all tax rates that attain that minimum, levy the lowest one.

3.2.2 Follower (supply chain operator)

Given the announced tax rate, the operator minimizes total operating cost plus tax payments by jointly deciding facility opening and installed capacity, technology installation, transportation modes and fleet, forward and reverse product flows, production, remanufacturing, and inventory.

4 Mathematical Model

We formulate the CLSC design problem with the carbon tax as a leader decision variable, as a bilevel mixed-integer programme (BL-MIP). A regulator (leader) selects a carbon-tax index; a supply chain operator (follower) then optimises network design and operations in response. Section 4.1 states the modelling assumptions. Section 4.2 establishes notation and auxiliary expressions. Section 4.3 presents the compact bilevel programme and a complexity result. Sections 4.4 and 4.5 develop the regulator's and operator's problems in detail.

4.1 Assumptions

The model rests on the following assumptions, grouped by the aspect of the problem they govern.

Data and cost structure.

A1. Demand d_{rpt} and return-rate bounds $[\underline{\eta}_{rpt}, \overline{\eta}_{rpt}]$ are known with certainty over the entire planning horizon $\mathcal{T}$.

A2. All unit economic costs (production, remanufacturing, transport, inventory holding, and material acquisition) are constant across periods and scale linearly with quantities.

A3. All emission factors are constant and technology- or mode-specific; emissions scale linearly with processed output, transported mass, and distance.

A4. Transportation lead times are absorbed within a single planning period.

Operational constraints.

A5. Every market demand is met exactly in each period; no inventory is held at markets.

A6. Hub terminals operate as pure cross-docking facilities: all items arriving at a hub in period t depart in the same period, with no on-site storage. Following Mota et al. [2018], hubs are treated as outsourced terminal-access points; selecting a hub incurs fixed access costs and flow-dependent handling costs, but no facility-construction or opening emissions.

A7. For every admissible index $\hat{\tau} \in \{\underline{\hat{\tau}}, \ldots, \overline{\hat{\tau}}\}$, the follower's feasible region $\mathcal{X}$ is non-empty. This can be verified a priori by checking that the network has sufficient installed capacity to cover maximum aggregate demand.

Bilevel game structure.

A8. The regulator announces the carbon-tax index $\hat{\tau}$ before the supply chain operator makes any facility, technology, or operational decision; the operator observes $\hat{\tau}$ with certainty and responds rationally (*open-loop Stackelberg game*).

A9. Facility-opening, supplier and hub selection, and technology-installation decisions are made at the start of the planning horizon and are irrevocable; no opening, closure, selection change, or capacity expansion occurs mid-horizon.

A10. The supply chain operator is a single entity that centrally coordinates all network design and operational decisions across all echelons.

A11. When the follower's optimal response is non-unique, the leader selects the response with the lowest total emissions (*optimistic bilevel convention*).

4.2 Notation

The following notation is used throughout the formulation. All sets, parameters, and decision variables are summarised in Tables 1 to 6.

Table 1. Sets and indices.

Set	Index	Description
Network nodes		
$\mathcal{S}$	s	Suppliers
$\mathcal{F}$	f	Factories
$\mathcal{W}$	w	Warehouses
$\mathcal{R}$	r	Markets (retailers)
$\mathcal{H}^{\mathrm{a}}, \mathcal{H}^{\mathrm{s}}$	h	Airports; seaports
$\mathcal{H}$	h	Hub terminals, $\mathcal{H} \coloneqq \mathcal{H}^{\mathrm{a}} \cup \mathcal{H}^{\mathrm{s}}$
$\mathcal{N}$	i, j	All nodes, $\mathcal{N} \coloneqq \mathcal{S} \cup \mathcal{F} \cup \mathcal{W} \cup \mathcal{R} \cup \mathcal{H}$
$\mathcal{N}^{\mathrm{o}}$	i	Openable nodes, $\mathcal{N}^{\mathrm{o}} \coloneqq \mathcal{S} \cup \mathcal{F} \cup \mathcal{W} \cup \mathcal{H}$
$\mathcal{N}^{\mathrm{v}}$	i	Inventory nodes, $\mathcal{N}^{\mathrm{v}} \coloneqq \mathcal{W}$
Items		
$\mathcal{U}, \mathcal{P}, \mathcal{K}$	u, p, k	Raw materials; finished products; recoverable items
$\mathcal{Q}$	q	All transportable items, $\mathcal{Q} \coloneqq \mathcal{U} \cup \mathcal{P} \cup \mathcal{K}$
Technologies		
$\mathcal{G}, \mathcal{L}$	g, ℓ	Production and remanufacturing technologies
Ω^{p}	(p, g)	Product-technology pairs for production, $\Omega^{\mathrm{p}} \subseteq \mathcal{P} \times \mathcal{G}$
Ω^{r}	(p, ℓ)	Product-technology pairs for remanufacturing, $\Omega^{\mathrm{r}} \subseteq \mathcal{P} \times \mathcal{L}$
Transport modes		
$\mathcal{M}^{\mathrm{rd}}$	m	Company-owned trucks (road mode)
$\mathcal{M}^{\mathrm{a}}, \mathcal{M}^{\mathrm{s}}$	m	Air; sea transfer modes
$\mathcal{M}^{\mathrm{out}}$	m	Outsourced modes, $\mathcal{M}^{\mathrm{out}} \coloneqq \mathcal{M}^{\mathrm{a}} \cup \mathcal{M}^{\mathrm{s}}$
$\mathcal{M}$	m	All transport modes, $\mathcal{M} \coloneqq \mathcal{M}^{\mathrm{rd}} \cup \mathcal{M}^{\mathrm{out}}$
Arcs		
$\mathcal{A}$	(i, j)	Admissible arcs
$\mathcal{A}_q$	(i, j)	Admissible arcs for item q
$\mathcal{M}_{ij}$	m	Modes permitted on arc (i, j)
Time		
$\mathcal{T}$	t	Planning periods, $\mathcal{T} \coloneqq \{1, \ldots, T\}$

Table 2. Economic parameters.

Parameter	Description	Units
F_i	Fixed cost of selecting supplier $i \in \mathcal{S}$	USD
c_i^{area}	Installation cost per unit of floor area at facility $i \in \mathcal{F} \cup \mathcal{W}$	USD/m^2
F_h	Per-period access (contract) cost of hub $h \in \mathcal{H}$	USD/period
$C_g^{\mathrm{p}},\ C_\ell^{\mathrm{r}}$	Installation cost of production/remanufacturing technology	USD
$c_g^{\mathrm{p}},\ c_\ell^{\mathrm{r}}$	Unit operating cost of production/remanufacturing	USD/unit
c_{su}^{u}	Unit acquisition cost of raw material u from supplier s	USD/unit
c_k^{k}	Unit acquisition cost of recovered item k	USD/unit
c_p^{h}	Unit holding cost of finished product p per period (warehouses only)	USD/(unit·period)
c_m^{tr}	Variable outsourced transport cost, $m \in \mathcal{M}^{\mathrm{out}}$	USD/(kg·km)
c_m^{veh}	Road transport cost per vehicle-km, $m \in \mathcal{M}^{\mathrm{rd}}$	USD/(veh·km)
c_m^{hub}	Unit handling cost at a hub terminal for mode m	USD/kg
F_m^{m}	Vehicle investment cost, road mode $m \in \mathcal{M}^{\mathrm{rd}}$	USD

Table 3. Capacity, resource, and fleet parameters.

Parameter	Description
$\underline{S}_{su}$	Min. per-period supply of raw material u from supplier s
$\overline{S}_{su}$	Max. per-period supply of raw material u from supplier s
$\underline{Q}^{\mathrm{p}}_{g}$	Min. aggregate output of production technology g per period
$\overline{Q}^{\mathrm{p}}_{g}$	Max. aggregate output of production technology g per period
$\underline{Q}^{\mathrm{r}}_{\ell}$	Min. aggregate output of remanufacturing technology ℓ per period
$\overline{Q}^{\mathrm{r}}_{\ell}$	Max. aggregate output of remanufacturing technology ℓ per period
$\underline{I}_{pi}$	Min. inventory of finished product p at warehouse $i \in \mathcal{N}^{\mathrm{v}}$
$\overline{I}_{pi}$	Max. inventory of finished product p at warehouse $i \in \mathcal{N}^{\mathrm{v}}$
I^{0}_{pi}	Initial inventory ($t = 0$) of finished product p at warehouse $i \in \mathcal{N}^{\mathrm{v}}$
$\underline{A}_{i}$	Min. installation area of facility $i \in \mathcal{F} \cup \mathcal{W}$
$\overline{A}_{i}$	Max. installation area of facility $i \in \mathcal{F} \cup \mathcal{W}$
$\overline{B}_{m}$	Contracted per-arc-period capacity of mode $m \in \mathcal{M}^{\mathrm{out}}$
$\underline{W}_{m}$	Min. arc-period weight when mode m is active
$\overline{W}_{m}$	Max. arc-period weight when mode m is active
M^{trip}_{m}	Big-M bound on trips by mode m per arc-period
r_m	Return-distance factor ($r_m = 1$ for outsourced modes)
v_m, H_m	Average speed, driving-hour limit (road)
$\overline{V}$	Max. total investment in company-owned vehicles
apu_q	Storage area per unit of item $q \in \mathcal{Q}$ held in end-of-period inventory; only $q \in \mathcal{P}$ at warehouses carries nonzero stock
$apur_p$	Handling area per unit of period throughput of finished product $p \in \mathcal{P}$, entering the flow-based component of (4.5)

Table 4. Demand, return, and conversion parameters.

Parameter	Description	Units
α_{upg}	Raw-material input ratio for u per unit of p via g	unit/unit
β_{kp}	Recovered-item input ratio for k per remanufactured unit of p	unit/unit
γ_{kp}	Yield of recovered item k per returned unit of p	unit/unit
ω_q	Unit weight of item q	kg
d_{rpt}	Demand for product p at market r in period t	units/period
$\underline{\eta}_{rpt}$	Min. return-rate fraction for p at market r in period t	–
$\overline{\eta}_{rpt}$	Max. return-rate fraction for p at market r in period t	–
θ_p	Integer return lag of product p	periods
δ_{ij}	Distance along arc (i, j)	km

Table 5. Emission-intensity and leader-policy parameters.

Parameter	Description	Units
$e^{\mathrm{o,f}}_{i}$	Opening emission intensity for facility $i \in \mathcal{F} \cup \mathcal{W}$	$\mathrm{kg\,CO_2e/m^2}$
$e^{\mathrm{p}}_{g},\ e^{\mathrm{r}}_{\ell}$	Emission intensity per kg produced/remanufactured	$\mathrm{kg\,CO_2e/kg}$
e^{u}_{su}	Emission intensity of raw material u from supplier s	$\mathrm{kg\,CO_2e/kg}$
e^{tr}_{m}	Transport emission intensity of mode m	$\mathrm{kg\,CO_2e/(kg{\cdot}km)}$
$\underline{\hat{\tau}}, \overline{\hat{\tau}}$	Lower/upper bounds on the leader's carbon-tax index	–

Table 6. Follower decision variables.

Variable	Domain	Description
y_i	$\{0,1\}$	1 if node $i \in \mathcal{N}^{\mathrm{o}}$ is activated (opened, if $i \in \mathcal{F} \cup \mathcal{W}$; selected, if $i \in \mathcal{S} \cup \mathcal{H}$)
YC_i	$\mathbb{R}_{\geq 0}$	Installed floor area at facility $i \in \mathcal{F} \cup \mathcal{W}$ (m^2)
$\lambda^{\mathrm{p}}_{fg}$	$\{0,1\}$	1 if production technology g is installed at f (shared across products)
$\lambda^{\mathrm{r}}_{f\ell}$	$\{0,1\}$	1 if remanufacturing technology ℓ is installed at f (shared across products)
$a_{f\ell t}$	$\{0,1\}$	1 if installed technology ℓ is operated at f in period t
σ_{sut}	$\{0,1\}$	1 if supplier s ships raw material u in period t
ζ_{ijmt}	$\{0,1\}$	1 if mode m is active on arc (i,j) in period t
κ_{mt}	$\{0,1\}$	1 if outsourced capacity for mode m is contracted in period t
Q_{ijmt}	$\mathbb{Z}_{\geq 0}$	Trips on arc (i,j) by road mode $m \in \mathcal{M}^{\mathrm{rd}}$ in period t
K_{im}	$\mathbb{Z}_{\geq 0}$	Company-owned vehicles of road mode m assigned to node i
q^{p}_{pfgt}	$\mathbb{R}_{\geq 0}$	Production of p at f via technology g in t, $(p,g) \in \Omega^{\mathrm{p}}$
$q^{\mathrm{r}}_{pf\ell t}$	$\mathbb{R}_{\geq 0}$	Remanufacturing of p at f via ℓ in t, $(p,\ell) \in \Omega^{\mathrm{r}}$
s^{inv}_{pit}	$\mathbb{R}_{\geq 0}$	Inventory of finished product p at warehouse $i \in \mathcal{N}^{\mathrm{v}}$ in period t
x_{qijmt}	$\mathbb{R}_{\geq 0}$	Flow of item q on arc (i,j) by mode m in period t

4.2.1 Sets and indices

Table 1 lists the node, item, technology, transport, arc, and planning-period sets. The set of all transportable items is $\mathcal{Q} = \mathcal{U} \cup \mathcal{P} \cup \mathcal{K}$ and the set of all nodes is $\mathcal{N} = \mathcal{S} \cup \mathcal{F} \cup \mathcal{W} \cup \mathcal{R} \cup \mathcal{H}$. Openable nodes are $\mathcal{N}^{\mathrm{o}} = \mathcal{S} \cup \mathcal{F} \cup \mathcal{W} \cup \mathcal{H}$; inventory-holding nodes are $\mathcal{N}^{\mathrm{v}} = \mathcal{W}$ – factories carry no inventory of their own (Section 4.5.2), so finished-product stock is held only at warehouses. Transport partition as $\mathcal{M} = \mathcal{M}^{\mathrm{rd}} \cup \mathcal{M}^{\mathrm{out}}$, where $\mathcal{M}^{\mathrm{out}} = \mathcal{M}^{\mathrm{a}} \cup \mathcal{M}^{\mathrm{s}}$ collects the outsourced non-road modes. Instance files additionally carry a set labelled I, kept equal to $\mathcal{N}$ for backward compatibility with an earlier node-indexing convention; it is validated against $\mathcal{N}$ on load but is not an independent model object, and the implementation reads only $\mathcal{N}$.

4.2.2 Parameters

Table 2 lists economic cost parameters; Table 3 lists capacity, resource, and fleet parameters; Table 4 lists demand, return, and conversion parameters; Table 5 lists emission intensities and the leader's policy bounds. The set $\mathcal{M}_{ij} \subseteq \mathcal{M}$ denotes the modes permitted on arc (i,j), and δ_{ij} its distance. $\mathcal{M}_{ij}$ is general per-arc instance data; the data generator and implementation realise it via a fixed topological rule – air-air arcs admit only $\mathcal{M}^{\mathrm{a}}$, sea-sea arcs only $\mathcal{M}^{\mathrm{s}}$, and every other arc admits the road-family modes $\mathcal{M}^{\mathrm{rd}}$ – which is sufficient for every benchmark topology used in this paper but is not the most general case the notation $\mathcal{M}_{ij}$ permits. The indicator

$$\xi^{\mathrm{o}}_{ij} = \mathbf{1}[i \in \mathcal{H}], \qquad \xi^{\mathrm{d}}_{ij} = \mathbf{1}[j \in \mathcal{H}]$$

flags whether the origin or destination of arc (i,j) is a hub terminal, used to assess handling charges. Compatible product–technology pairs are collected in $\Omega^{\mathrm{p}} \subseteq \mathcal{P} \times \mathcal{G}$ (production) and $\Omega^{\mathrm{r}} \subseteq \mathcal{P} \times \mathcal{L}$ (remanufacturing).

4.2.3 Decision variables

Leader decision variable. The regulator's sole decision is the integer carbon-tax index

$$\hat{\tau} \in \mathcal{T}^{\tau} \coloneqq \{\underline{\hat{\tau}}, \underline{\hat{\tau}}+1, \ldots, \overline{\hat{\tau}}\} \subset \mathbb{Z}, \tag{4.1}$$

which induces the carbon-tax rate $\tau = \hat{\tau}/100$ (USD/kg CO_2e). We work with the index $\hat{\tau}$ rather than the rate τ directly because it keeps the leader's domain finite and discrete, enabling the exact DBP of Section 5.

Follower decision variables. Given $\hat{\tau}$, the follower chooses the vector

$$\mathbf{z} = (y,\, YC,\, \lambda^{\mathrm{p}},\, \lambda^{\mathrm{r}},\, a,\, \sigma,\, \zeta,\, \kappa,\, Q,\, K,\, q^{\mathrm{p}},\, q^{\mathrm{r}},\, s^{\mathrm{inv}},\, x).$$

Table 6 lists all component variables and their domains.

To decouple the algebraic model from the physical network, the admissible flow-tuple set is

$$\Phi \coloneqq \{(q,i,j) \,:\, q \in \mathcal{Q},\, (i,j) \in \mathcal{A}_q\}, \tag{4.2}$$

which admits the partition $\Phi = \Phi^{\mathrm{u}} \cup \Phi^{\mathrm{p}} \cup \Phi^{\mathrm{k}}$ according to $q \in \mathcal{U}$, $q \in \mathcal{P}$, or $q \in \mathcal{K}$.

4.2.4 Auxiliary expressions

The following shorthand expressions are used throughout the formulation to keep the notation compact. For each item $q \in \mathcal{Q}$, node $i \in \mathcal{N}$, and period $t \in \mathcal{T}$, the aggregate inflow and outflow at node i are

$$X^{\mathrm{in}}_{qit} \coloneqq \sum_{j:(j,i)\in\mathcal{A}_q} \sum_{m\in\mathcal{M}_{ji}} x_{qjimt}, \qquad X^{\mathrm{out}}_{qit} \coloneqq \sum_{j:(i,j)\in\mathcal{A}_q} \sum_{m\in\mathcal{M}_{ij}} x_{qijmt}. \tag{4.3}$$

To avoid case distinctions between openable and non-openable nodes in the flow constraints,

$$\bar{y}_i \coloneqq \begin{cases} y_i, & i \in \mathcal{N}^{\mathrm{o}}, \\ 1, & i \in \mathcal{R}. \end{cases} \tag{4.4}$$

The used floor area at facility $i \in \mathcal{F} \cup \mathcal{W}$ in period t combines a flow-based handling component, present at both factories and warehouses, with a stock-based storage component, present only at warehouses (factories carry no inventory of any kind, Section 4.5.2):

$$YCT_{it} \coloneqq \begin{cases} \displaystyle\sum_{p\in\mathcal{P}} apur_p\, X^{\mathrm{out}}_{pit}, & i \in \mathcal{F}, \\ \displaystyle\sum_{p\in\mathcal{P}} apur_p\, X^{\mathrm{in}}_{pit} + \sum_{p\in\mathcal{P}} apu_p\, s^{\mathrm{inv}}_{pit}, & i \in \mathcal{W}, \end{cases} \qquad t \in \mathcal{T}. \tag{4.5}$$

Factories carry no inventory of any kind (Section 4.5.2), so their floor-area footprint has no stock component; their handling space is driven entirely by per-period throughput. Warehouses carry

both: incoming-flow handling space and finished-product stock space, following Mota et al. [2018]'s general facility-area expression (their Eq. 16), adapted to this project's flow-direction convention — see Theorem 4.2.

4.3 Bilevel Programme

The CLSC design problem with the carbon tax as a leader decision variable is modelled as the following bilevel programme. The leader announces a carbon-tax index $\hat{\tau}$; the follower observes $\hat{\tau}$ and selects the cost-minimising supply chain design and operations. Denoting the follower's optimal-response correspondence by $\Psi(\hat{\tau})$ and its feasible region by $\mathcal{X}$ (defined in Section 4.5), the complete programme is

$$\text{(BL)} \quad \operatorname*{lex\,min}_{\hat{\tau},\,\mathbf{z}} \quad \big(E(\mathbf{z}),\ \hat{\tau}\big) \tag{4.6a}$$

$$\text{s.t.} \quad \hat{\tau} \in \mathcal{T}^{\tau}, \tag{4.6b}$$

$$\mathbf{z} \in \Psi(\hat{\tau}), \tag{4.6c}$$

where $\Psi(\hat{\tau})$ is defined in (4.8) once the follower's objective has been introduced in Section 4.5.1. Sections 4.4 and 4.5 present the two levels in detail.

Bilevel programmes with integer lower-level variables are NP-hard in general [Moore and Bard, 1990, Jeroslow, 1985]; problem (4.6) inherits this hardness, since the follower's problem (4.7) is itself a MILP for every fixed $\hat{\tau}$.

4.4 Regulator's Problem

4.4.1 Objective function

The regulator is a *lexicographic minimiser*: it first minimises the total emissions $E(\mathbf{z})$ induced by the follower's response and, among all tax indices that attain that minimum, selects the smallest one (see (4.6a)).

Remark 4.1 (Lexicographic structure). The lexicographic objective (4.6a) minimises emissions unconditionally, with the tax index acting solely as a tiebreaker. This secondary criterion encodes a *least-intervention principle*: the regulator does not levy a tax heavier than the emissions target requires. Section 6.2 shows this matters when the emission–tax response saturates and the primary minimiser is attained across a wide band of tax indices.

4.4.2 Feasibility constraints

Two constraints define the leader's feasible set. Constraint (4.6b) restricts the tax to the admissible grid $\mathcal{T}^{\tau}$ (Section 4.2.3). Constraint (4.6c) imposes bilevel feasibility: the supply chain operator's response must be optimal given the announced tax. Under the optimistic tie-break of Assumption A11, the leader selects, among all optimal follower responses, the one with the lowest emissions.

4.5 Supply Chain Operator's Problem

Given the tax index $\hat{\tau} \in \mathcal{T}^{\tau}$ announced by the regulator, the supply chain operator solves a mixed-integer programme whose feasible region $\mathcal{X}$ is defined by constraints (C1)–(C10) in Sections 4.5.2 to 4.5.11. The objective, detailed in Section 4.5.1, combines total cost and carbon-tax payment.

4.5.1 Objective function

Given $\hat{\tau}$, the operator minimises the sum of its total cost $C(\mathbf{z})$ and carbon-tax payment:

$$\text{(FL)} \quad \min_{\mathbf{z}\in\mathcal{X}} \; C(\mathbf{z}) + \tfrac{\hat{\tau}}{100}\, E(\mathbf{z}), \tag{4.7}$$

with optimal-response correspondence

$$\Psi(\hat{\tau}) \coloneqq \arg\min\nolimits_{\mathbf{z}\in\mathcal{X}} \Big\{ C(\mathbf{z}) + \tfrac{\hat{\tau}}{100}\, E(\mathbf{z}) \Big\}. \tag{4.8}$$

The operator's total cost is given by

$$\begin{aligned} C(\mathbf{z}) = {} & \sum_{i\in\mathcal{S}} F_i\, y_i + \sum_{i\in\mathcal{F}\cup\mathcal{W}} c_i^{\text{area}}\, YC_i + \sum_{f\in\mathcal{F}} \Big(\sum_{g\in\mathcal{G}} C_g^{\text{p}}\, \lambda_{fg}^{\text{p}} + \sum_{\ell\in\mathcal{L}} C_\ell^{\text{r}}\, \lambda_{f\ell}^{\text{r}} \Big) \\ & + \sum_{i\in\mathcal{N}} \sum_{m\in\mathcal{M}^{\text{rd}}} F_m^{\text{m}}\, K_{im} + \sum_{h\in\mathcal{H}} \sum_{t\in\mathcal{T}} F_h\, y_h \\ & + \sum_{(p,g)\in\Omega^{\text{p}}} \sum_{f,t} c_g^{\text{p}}\, q_{pfgt}^{\text{p}} + \sum_{(p,\ell)\in\Omega^{\text{r}}} \sum_{f,t} c_\ell^{\text{r}}\, q_{pf\ell t}^{\text{r}} \\ & + \sum_{p\in\mathcal{P}} \sum_{i\in\mathcal{N}^{\text{v}}} \sum_{t} c_p^{\text{h}}\, s_{pit}^{\text{inv}} \\ & + \sum_{u\in\mathcal{U}} \sum_{s\in\mathcal{S}} \sum_{t} c_{su}^{\text{u}}\, X_{ust}^{\text{out}} + \sum_{k\in\mathcal{K}} \sum_{r\in\mathcal{R}} \sum_{t} c_k^{\text{k}}\, X_{krt}^{\text{out}} \\ & + \sum_{(i,j)\in\mathcal{A}} \sum_{m\in\mathcal{M}_{ij}\cap\mathcal{M}^{\text{rd}}} \sum_{t} c_m^{\text{veh}}\, r_m\, \delta_{ij}\, Q_{ijmt} + \sum_{(q,i,j)\in\Phi} \sum_{m\in\mathcal{M}_{ij}\setminus\mathcal{M}^{\text{rd}}} \sum_{t} c_m^{\text{tr}}\, \delta_{ij}\, \omega_q\, x_{qijmt} \\ & + \sum_{(q,i,j)\in\Phi} \sum_{m\in\mathcal{M}_{ij}} \sum_{t} c_m^{\text{hub}} (\xi_{ij}^{\text{o}} + \xi_{ij}^{\text{d}}) \omega_q\, x_{qijmt}, \end{aligned} \tag{4.9}$$

The first term is the fixed cost of selecting a supplier, plus the area-proportional installation cost c_i^{area} charged on the floor area YC_i installed at an opened factory or warehouse – there is no separate fixed cost for opening a factory or warehouse, since $YC_i \geq \underline{A}_i\, y_i$ already forces a nonzero installation cost whenever $y_i = 1$ (Equation (4.14a)) – plus the cost of installing a production or remanufacturing technology at each factory. The second term is the road fleet investment cost and the hub access fee, the latter charged every period a hub is selected. The third term is the variable cost of production and remanufacturing, and the fourth is the inventory-holding cost on finished-product stock at warehouses – factories and recovered items are never held in inventory (Section 4.5.2), so neither carries a holding-cost term. The fifth term is the acquisition cost of raw materials and recovered items. The sixth term is transport: road transport is charged per trip, scaled by the return factor r_m for the empty return leg,

while outsourced transport is charged per kilogram-kilometre shipped. The seventh and final term is the hub-handling charge, applied once for each hub endpoint of an arc.

Total emissions are given by

$$\begin{aligned} E(\mathbf{z}) = & \sum_{i \in \mathcal{F} \cup \mathcal{W}} e_i^{\mathrm{o,f}}\, YC_i \\ & + \sum_{(p,g) \in \Omega^{\mathrm{p}}} \sum_{f,t} e_g^{\mathrm{p}}\, \omega_p\, q^{\mathrm{p}}_{pfgt} + \sum_{(p,\ell) \in \Omega^{\mathrm{r}}} \sum_{f,t} e_\ell^{\mathrm{r}}\, \omega_p\, q^{\mathrm{r}}_{pf\ell t} \\ & + \sum_{u \in \mathcal{U}} \sum_{s \in \mathcal{S}} \sum_{t} e^{\mathrm{u}}_{su}\, \omega_u\, X^{\mathrm{out}}_{ust} \\ & + \sum_{(q,i,j) \in \Phi} \sum_{m \in \mathcal{M}_{ij}} \sum_{t} e^{\mathrm{tr}}_m\, r_m\, \omega_q\, \delta_{ij}\, x_{qijmt}. \end{aligned} \tag{4.10}$$

The first term is the opening emission: an amount proportional to the installed area YC_i for an opened factory or warehouse – suppliers and hubs carry no opening emission of their own; a supplier's emission footprint is captured entirely by the raw-material acquisition term below. The second term is the production and remanufacturing emission, per kilogram of output. The third term is the raw-material acquisition emission, per kilogram shipped. The fourth and final term is transport, scaled by the same return factor r_m used in the transport cost (Equation (4.9)): the empty return leg of a road delivery still burns fuel and emits CO_2, even though it carries no cargo; $r_m = 1$ for outsourced modes, which have no return leg attributable to the follower.

4.5.2 (C1) Flow conservation and inventory balance

These constraints enforce material balance at every node type in every period. Finished-product inventory follows a standard stock-balance recurrence at warehouses – factories carry no inventory of their own and operate as pure flow-through nodes for finished products, mirroring the hub cross-docking pattern below; recovered items are likewise never buffered, at warehouses or anywhere else, and pass through warehouses as pure flow-through too; markets satisfy demand exactly with no on-site inventory; and hub terminals operate as pure cross-docking nodes:

$$X^{\mathrm{in}}_{pft} + \sum_{g:(p,g) \in \Omega^{\mathrm{p}}} q^{\mathrm{p}}_{pfgt} + \sum_{\ell:(p,\ell) \in \Omega^{\mathrm{r}}} q^{\mathrm{r}}_{pf\ell t} = X^{\mathrm{out}}_{pft}, \qquad \forall p \in \mathcal{P},\ f \in \mathcal{F},\ t \in \mathcal{T}, \tag{4.11a}$$

$$s^{\mathrm{inv}}_{pwt} = s^{\mathrm{inv}}_{pw,t-1} + X^{\mathrm{in}}_{pwt} - X^{\mathrm{out}}_{pwt}, \forall p \in \mathcal{P},\ w \in \mathcal{W},\ t \in \mathcal{T}, \tag{4.11b}$$

$$X^{\mathrm{in}}_{kwt} = X^{\mathrm{out}}_{kwt}, \qquad \forall k \in \mathcal{K},\ w \in \mathcal{W},\ t \in \mathcal{T}, \tag{4.11c}$$

$$X^{\mathrm{in}}_{prt} = d_{rpt}, \qquad \forall p \in \mathcal{P},\ r \in \mathcal{R},\ t \in \mathcal{T}, \tag{4.11d}$$

$$X^{\mathrm{in}}_{qht} = X^{\mathrm{out}}_{qht}, \qquad \forall q \in \mathcal{Q},\ h \in \mathcal{H},\ t \in \mathcal{T}, \tag{4.11e}$$

with boundary condition $s^{\text{inv}}_{pi,0} = I^0_{pi}$ for all $p \in \mathcal{P}$, $i \in \mathcal{N}^{\text{v}}$.

4.5.3 (C2) Reverse logistics and returns

Return flows are governed by product-specific collection rates and lag parameters. The volume of recoverable item k collected at market r in period t is bounded above and below by fractions of lagged demand (with $d_{rp,t-\theta_p} \coloneqq 0$ when $t - \theta_p < 1$). Every recovered unit entering a factory is consumed entirely by remanufacturing, closing the material loop:

$$\sum_{p \in \mathcal{P}} \gamma_{kp}\, \underline{\eta}_{rpt}\, d_{rp,t-\theta_p} \le X^{\text{out}}_{krt} \le \sum_{p \in \mathcal{P}} \gamma_{kp}\, \overline{\eta}_{rpt}\, d_{rp,t-\theta_p}, \forall k \in \mathcal{K},\, r \in \mathcal{R},\, t \in \mathcal{T}, \tag{4.12a}$$

$$X^{\text{in}}_{kft} = \sum_{(p,\ell) \in \Omega^{\text{r}}} \beta_{kp}\, q^{\text{r}}_{pf\ell t}, \qquad \forall k \in \mathcal{K},\, f \in \mathcal{F},\, t \in \mathcal{T}. \tag{4.12b}$$

4.5.4 (C3) Raw material supply

A supplier can ship raw material u in period t only if it is both open and activated via the binary σ_{sut}. The shipped quantity is then bounded within the supplier's per-period capacity, and the total raw-material inflow at each factory in period t must exactly satisfy production stoichiometry:

$$\underline{S}_{su}\, \sigma_{sut} \le X^{\text{out}}_{ust} \le \overline{S}_{su}\, \sigma_{sut}, \qquad \forall s \in \mathcal{S},\, u \in \mathcal{U},\, t \in \mathcal{T}, \tag{4.13a}$$

$$\sigma_{sut} \le y_s, \qquad \forall s \in \mathcal{S},\, u \in \mathcal{U},\, t \in \mathcal{T}, \tag{4.13b}$$

$$X^{\text{in}}_{uft} = \sum_{(p,g) \in \Omega^{\text{p}}} \alpha_{upg}\, q^{\text{p}}_{pfgt}, \forall u \in \mathcal{U},\, f \in \mathcal{F},\, t \in \mathcal{T}. \tag{4.13c}$$

4.5.5 (C4) Facility opening and area capacity

The floor area installed at each factory and warehouse is a continuous decision that is active only when the facility is open. The installed area is bounded by facility-specific limits conditional on opening, and the used floor area YCT_{it} (Equation (4.5)) – combining handling space for period throughput and, at warehouses, storage space for end-of-period inventory – must not exceed the installed area:

$$\underline{A}_i\, y_i \le YC_i \le \overline{A}_i\, y_i, \forall i \in \mathcal{F} \cup \mathcal{W}, \tag{4.14a}$$

$$YCT_{it} \le YC_i, \forall i \in \mathcal{F} \cup \mathcal{W},\, t \in \mathcal{T}. \tag{4.14b}$$

Remark 4.2 (Flow-direction adaptation of the floor-area expression). Mota et al. [2018]'s general facility-area expression (their Eq. 16) combines a flow-based handling term and a stock-based storage term, $\sum_{m,a,j} apur_m X_{majit} + \sum_m apu_m S_{mit}$, where X denotes flow *into* node i in their generic indexing convention. This project adapts the flow-based term to its own flow-direction convention rather than copying X_{majit} verbatim. At a factory, $X^{\text{in}}_{pft} \equiv 0$ structurally under the greenfield network-design convention of Theorem 4.5 — no arc routes finished product *into* a factory — so a literal inbound-flow term would always vanish and defeat the purpose of modelling factory handling space at all. Equation (4.5) therefore uses

the economically meaningful throughput quantity at a factory, the outbound flow X^{out}_{pft}, which by Theorem 4.5 equals $q^{\text{p}}_{pfgt} + q^{\text{r}}_{pf\ell t}$ under that convention but remains well defined as a direct flow quantity even if it did not. At a warehouse, both X^{in}_{pwt} and X^{out}_{pwt} are generally nonzero and differ only by the stock change (4.11b), so either is a defensible throughput proxy; (4.5) retains the inbound flow X^{in}_{pwt} as the more literal mirror of Mota's own indexing convention, representing the receiving/staging space needed for incoming goods, kept distinct from the already-present stock term.

4.5.6 (C5) Inventory bounds

Physical capacity and handling requirements impose per-product bounds on end-of-period inventory. These bounds are active only when the facility is open, linking the location and inventory decisions:

$$\underline{I}_{pi}\, y_i \le s^{\text{inv}}_{pit} \le \overline{I}_{pi}\, y_i, \forall p \in \mathcal{P},\, i \in \mathcal{N}^{\text{v}},\, t \in \mathcal{T}. \tag{4.15a}$$

4.5.7 (C6) Technology selection and operation

Production and remanufacturing technologies are selected at the factory level and shared across all compatible products in set Ω^{p} or Ω^{r}. Aggregate throughput per installed technology is bounded in each period; remanufacturing operation in any period requires prior installation; all technology installations require the factory to be open; and at most one technology per product may be installed at each factory:

$$\underline{Q}^{\text{p}}_{g}\, \lambda^{\text{p}}_{fg} \le \sum_{p:(p,g)\in\Omega^{\text{p}}} q^{\text{p}}_{pfgt} \le \overline{Q}^{\text{p}}_{g}\, \lambda^{\text{p}}_{fg}, \forall f \in \mathcal{F},\, g \in \mathcal{G},\, t \in \mathcal{T}, \tag{4.16a}$$

$$\underline{Q}^{\text{r}}_{\ell}\, a_{f\ell t} \le \sum_{p:(p,\ell)\in\Omega^{\text{r}}} q^{\text{r}}_{pf\ell t} \le \overline{Q}^{\text{r}}_{\ell}\, a_{f\ell t}, \forall f \in \mathcal{F},\, \ell \in \mathcal{L},\, t \in \mathcal{T}, \tag{4.16b}$$

$$a_{f\ell t} \le \lambda^{\text{r}}_{f\ell}, \qquad \forall f \in \mathcal{F},\, \ell \in \mathcal{L},\, t \in \mathcal{T}, \tag{4.16c}$$

$$\lambda^{\text{p}}_{fg},\ \lambda^{\text{r}}_{f\ell} \le y_f, \qquad \forall f \in \mathcal{F},\, g \in \mathcal{G},\, \ell \in \mathcal{L}, \tag{4.16d}$$

$$\sum_{g:(p,g)\in\Omega^{\text{p}}} \lambda^{\text{p}}_{fg} \le 1, \qquad \forall p \in \mathcal{P},\, f \in \mathcal{F}, \tag{4.16e}$$

$$\sum_{\ell:(p,\ell)\in\Omega^{\text{r}}} \lambda^{\text{r}}_{f\ell} \le 1, \qquad \forall p \in \mathcal{P},\, f \in \mathcal{F}. \tag{4.16f}$$

4.5.8 (C7) Transportation mode selection and shipments

Transport services are contracted on each arc per period through the binary mode-activation variable ζ_{ijmt}. For owned road modes, the total shipment weight must be consistent with the integer number of trips Q_{ijmt}. Outsourced non-road modes are activated directly by transported weight up to a contracted capacity, and at most one mode may operate on any

arc in any period:

$$\underline{W}_m\, Q_{ijmt} \le \sum_{q:(q,i,j)\in\Phi} \omega_q\, x_{qijmt} \le \overline{W}_m\, Q_{ijmt}, \forall (i,j)\in\mathcal{A},\ m\in\mathcal{M}_{ij}\cap\mathcal{M}^{\mathrm{rd}},\ t\in\mathcal{T}, \tag{4.17a}$$

$$\zeta_{ijmt} \le Q_{ijmt} \le M_m^{\mathrm{trip}}\,\zeta_{ijmt}, \forall (i,j)\in\mathcal{A},\ m\in\mathcal{M}_{ij}\cap\mathcal{M}^{\mathrm{rd}},\ t\in\mathcal{T}, \tag{4.17b}$$

$$\sum_{q:(q,i,j)\in\Phi} \omega_q\, x_{qijmt} \le \overline{B}_m\,\zeta_{ijmt}, \quad \forall (i,j)\in\mathcal{A},\ m\in\mathcal{M}_{ij}\cap\mathcal{M}^{\mathrm{out}},\ t\in\mathcal{T}, \tag{4.17c}$$

$$\sum_{m\in\mathcal{M}_{ij}} \zeta_{ijmt} \le 1, \qquad \forall (i,j)\in\mathcal{A},\ t\in\mathcal{T}. \tag{4.17d}$$

Remark 4.3 (Big-M implementation). M_m^{trip} in (4.17b) is presented above as a single constant per mode for notational economy. The implementation instead derives a tightened, arc- and node-specific bound from each arc's transport-capacity and node-throughput parameters (any value at least as large as the true bound preserves correctness, so a tighter, context-specific value is always admissible). An optional per-(node, mode) override, absent from the notation above, may be supplied through instance data to further tighten this bound when known a priori; see Section 5.5.

4.5.9 (C8) Fleet management and node activation

Owned road vehicles must be physically stationed at their dispatching node. A transport mode is usable only if both arc endpoints are active. Outsourced modes require a period-level contract. The fleet at each node is limited by driving-hour availability and by a total capital budget. Finally, every open facility must carry positive flow to prevent degenerate open-but-unused nodes:

$$\zeta_{ijmt} \le \bar{y}_i, \quad \zeta_{ijmt} \le \bar{y}_j, \qquad \forall (i,j)\in\mathcal{A},\ m\in\mathcal{M}_{ij},\ t\in\mathcal{T}, \tag{4.18a}$$

$$\zeta_{ijmt} \le \kappa_{mt}, \qquad \forall (i,j)\in\mathcal{A},\ m\in\mathcal{M}_{ij}\cap\mathcal{M}^{\mathrm{out}},\ t\in\mathcal{T}, \tag{4.18b}$$

$$\sum_{\substack{j:(i,j)\in\mathcal{A}\\ m\in\mathcal{M}_{ij}}} \frac{r_m\,\delta_{ij}}{v_m}\, Q_{ijmt} \le H_m\, K_{im}, \qquad \forall i\in\mathcal{N},\ m\in\mathcal{M}^{\mathrm{rd}},\ t\in\mathcal{T}, \tag{4.18c}$$

$$K_{im} \le M_m^{\mathrm{trip}}\,\bar{y}_i, \qquad \forall i\in\mathcal{N},\ m\in\mathcal{M}^{\mathrm{rd}}, \tag{4.18d}$$

$$K_{im} \le M_m^{\mathrm{trip}} \sum_{\substack{j:(i,j)\in\mathcal{A}\\ m\in\mathcal{M}_{ij}}} \sum_{t\in\mathcal{T}} Q_{ijmt}, \forall i\in\mathcal{N},\ m\in\mathcal{M}^{\mathrm{rd}}, \tag{4.18e}$$

$$\sum_{i\in\mathcal{N}} \sum_{m\in\mathcal{M}^{\mathrm{rd}}} F_m^{\mathrm{m}}\, K_{im} \le \overline{V}, \tag{4.18f}$$

$$\sum_{q\in\mathcal{Q}} \sum_{m\in\mathcal{M}} \sum_{t\in\mathcal{T}} \sum_{\substack{(i,j)\in\mathcal{A}\\ m\in\mathcal{M}_{ij}}} x_{qijmt} \ge y_i, \qquad \forall i\in\mathcal{N}^{\mathrm{o}}. \tag{4.18g}$$

Remark 4.4 (Degenerate transport-mode data). Equation (4.18c) presumes a positive mode speed v_m and a positive driving-hour allowance H_m. The implementation substitutes a trip-count bound (via the big-M of Theorem 4.3) whenever a mode's speed or driving-hour

parameter is non-positive, which cannot occur for any mode populated by the data generator of Section 5.5 but guards against division-by-zero on hand-edited or future instance data.

4.5.10 (C9) Intermodal hub routing

Intermodal terminals act as mode-switching cross-docking points. At each hub of type $\nu \in \{\mathrm{a}, \mathrm{s}\}$, the volume arriving by non-ν modes from non-ν sources must equal the volume departing to same-type hubs by ν-modes, and vice versa. Let $\overline{\mathcal{M}}_{ij}^{\nu} \coloneqq \mathcal{M}_{ij} \cap \mathcal{M}^{\nu}$ and $\mathcal{M}_{ij}^{-\nu} \coloneqq \mathcal{M}_{ij} \setminus \mathcal{M}^{\nu}$:

$$\sum_{\substack{i:(i,h)\in\mathcal{A}_q \\ i\notin\mathcal{H}^{\nu}}} \sum_{m\in\mathcal{M}_{ih}^{-\nu}} x_{qihmt} = \sum_{\substack{j:(h,j)\in\mathcal{A}_q \\ j\in\mathcal{H}^{\nu}}} \sum_{m\in\overline{\mathcal{M}}_{hj}^{\nu}} x_{qhjmt}, \forall q \in \mathcal{Q},\, h \in \mathcal{H}^{\nu},\, \nu,\, t \in \mathcal{T}, \tag{4.19a}$$

$$\sum_{\substack{i:(i,h)\in\mathcal{A}_q \\ i\in\mathcal{H}^{\nu}}} \sum_{m\in\overline{\mathcal{M}}_{ih}^{\nu}} x_{qihmt} = \sum_{\substack{j:(h,j)\in\mathcal{A}_q \\ j\notin\mathcal{H}^{\nu}}} \sum_{m\in\mathcal{M}_{hj}^{-\nu}} x_{qhjmt}, \forall q \in \mathcal{Q},\, h \in \mathcal{H}^{\nu},\, \nu,\, t \in \mathcal{T}. \tag{4.19b}$$

4.5.11 (C10) Valid inequalities

Every open factory must install at least one production or remanufacturing technology,

$$y_f \le \sum_{g\in\mathcal{G}} \lambda_{fg}^{\mathrm{p}} + \sum_{\ell\in\mathcal{L}} \lambda_{f\ell}^{\mathrm{r}}, \qquad \forall f \in \mathcal{F}, \tag{4.20}$$

A further candidate inequality would require aggregate installed capacity to cover aggregate demand in every period,

$$\sum_{f\in\mathcal{F}} \left(\sum_{g:(p,g)\in\Omega^{\mathrm{p}}} \overline{Q}_g^{\mathrm{p}}\, \lambda_{fg}^{\mathrm{p}} + \sum_{\ell:(p,\ell)\in\Omega^{\mathrm{r}}} \overline{Q}_\ell^{\mathrm{r}}\, \lambda_{f\ell}^{\mathrm{r}} \right) \ge \sum_{r\in\mathcal{R}} d_{rpt}, \qquad \forall p \in \mathcal{P},\, t \in \mathcal{T}, \tag{4.21}$$

but, unlike (4.20), this is *not* a valid inequality and is not part of the implemented formulation; see Theorem 4.5.

Remark 4.5 (Valid inequalities). Constraint (4.20) is implied whenever the flow topology never routes finished product *into* a factory — only the outbound arcs $\mathcal{A}_{fw}$ exist for $p \in \mathcal{P}$ (the standard convention for a greenfield network-design instance, and the one used throughout Section 6). Under that convention, $X_{pft}^{\mathrm{in}} = 0$ structurally, so the flow-balance equation (4.11a) collapses to $q_{pfgt}^{\mathrm{p}} + q_{pf\ell t}^{\mathrm{r}} = X_{pft}^{\mathrm{out}}$. A factory installing neither technology therefore has $X_{pft}^{\mathrm{out}} = 0$ directly, and the raw-material and recovered-item inflows that feed production and remanufacturing are likewise pinned to zero by the equalities (4.13c) and (4.12b). Such a factory carries zero flow of every item type, contradicting (4.18g) if it is open. Note that a single production technology alone — the original, narrower form of this inequality — is *not* implied: a factory engaged purely in remanufacturing satisfies (4.18g) via q^{r} alone, without ever installing λ^{p}. Constraint (4.20) is included in the implementation because it is not redundant in the LP relaxation; it tightens the linear-programming bound and reduces the enumeration effort when the DBP solves the fixed-tax follower subproblems in Section 5.

Constraint (4.21), by contrast, is *not* logically implied by (C1)–(C9) in general, because it ignores inventory banking: $\lambda_{fg}^{\mathrm{p}}$ and $\lambda_{f\ell}^{\mathrm{r}}$ are time-invariant, but a network can install less

capacity than a single period's peak demand and still satisfy that peak by overproducing in earlier, low-demand periods and drawing down s^{inv}_{pwt} at warehouses (Equation (4.11b)) when demand spikes, as long as the inventory bounds (4.15a) permit it. Any instance with a non-flat demand profile d_{rpt} and sufficient inventory headroom can have an optimal solution that violates (4.21) at the peak period while remaining feasible and cost/emissions-optimal under (C1)–(C9). Adding it as a hard constraint would therefore risk excluding genuinely optimal solutions rather than merely tightening the relaxation, so it is deliberately *not* included in the implemented formulation.

4.5.12 Domain constraints

$$y_i,\ \lambda^{\text{p}}_{fg},\ \lambda^{\text{r}}_{f\ell},\ a_{f\ell t},\ \sigma_{sut},\ \zeta_{ijmt},\ \kappa_{mt} \in \{0,1\}, \tag{4.22a}$$

$$Q_{ijmt},\ K_{im} \in \mathbb{Z}_{\geq 0}, \tag{4.22b}$$

$$YC_i,\ q^{\text{p}}_{pfgt},\ q^{\text{r}}_{pf\ell t},\ s^{\text{inv}}_{pit},\ x_{qijmt} \in \mathbb{R}_{\geq 0}. \tag{4.22c}$$

Constraints (4.11)–(4.22) define the follower's feasible region $\mathcal{X}$. Together with the auxiliary expressions (4.3)–(4.5), the objectives (4.9)–(4.10), the follower programme (4.7), and the bilevel programme (4.6), they constitute the BL-MIP solved in Section 5.

5 Solution Methodology

We develop a *Deterministic Bounding Procedure* (DBP) that solves problem (4.6) to global ϵ-optimality. The DBP maintains a lower bound LB on the leader's optimal emissions and an upper bound UB attained by a bilevel-feasible solution, tightening both iteratively until the optimality gap falls below a prescribed tolerance ϵ.

5.1 Lower-bound construction

A valid lower bound is obtained from a *relaxation master* that optimises over the same feasible region $\mathcal{X}$ as the follower (Section 4.5) – crucially, $\mathcal{X}$ does not depend on $\hat{\tau}$; only the follower's objective (4.7) does. Each time the follower is solved exactly at some tax index $\hat{\tau}^{(j)}$, obtaining the optimistic response $\mathbf{z}^{(j)} \in \Psi(\hat{\tau}^{(j)})$ with $c_j \coloneqq C(\mathbf{z}^{(j)})$ and $e_j \coloneqq E(\mathbf{z}^{(j)})$, the inequality

$$C(\mathbf{z}) + \tfrac{\hat{\tau}}{100}\, E(\mathbf{z}) \ \leq\ c_j + \tfrac{\hat{\tau}}{100}\, e_j \tag{5.1}$$

is valid for *every* $\hat{\tau} \in \mathcal{T}^{\tau}$ and every $\mathbf{z} \in \mathcal{X}$ (Theorem 5.1), because $\mathbf{z}^{(j)}$ remains a feasible – if not necessarily optimal – competitor in the follower's problem at any tax index. After k such evaluations have accumulated the cut set $\mathcal{C} = \{(\hat{\tau}^{(j)}, c_j, e_j)\}_{j<k}$, the relaxation master is

$$\text{LB}^{(k)} \ \coloneqq\ \min_{\hat{\tau}\in\mathcal{T}^{\tau},\,\mathbf{z}\in\mathcal{X}} \Big\{ E(\mathbf{z}) \ :\ (5.1) \text{ holds for every } (\hat{\tau}^{(j)}, c_j, e_j) \in \mathcal{C} \Big\}. \tag{5.2}$$

$\mathrm{LB}^{(k)}$ is non-decreasing in k, since each new cut only shrinks the master's feasible region, and bounds the leader's optimal emissions from below.

Lemma 5.1. *For every $k \geq 0$, $\mathrm{LB}^{(k)} \leq E(\mathbf{z}^*)$ for any optimal $(\hat{\tau}^*, \mathbf{z}^*)$ of (4.6).*

Proof. See Section A. □

5.2 Upper-bound construction

A valid upper bound is obtained by evaluating the follower exactly – to MILP optimality, with the optimistic tie-break of Assumption A11 – at the sequence of tax indices $\hat{\tau}^{(0)}, \hat{\tau}^{(1)}, \ldots$ proposed by the relaxation master of Section 5.1:

1. Solve the follower MIP (4.7) at $\hat{\tau} = \hat{\tau}^{(k)}$ to optimality, obtaining $\mathbf{z}^{(k)} \in \Psi(\hat{\tau}^{(k)})$.
2. Compute $E(\mathbf{z}^{(k)})$ via (4.10).
3. Set $\mathrm{UB}^{(k)} := \min\{\mathrm{UB}^{(k-1)}, E(\mathbf{z}^{(k)})\}$ (with $\mathrm{UB}^{(-1)} := +\infty$), updating the incumbent $(\hat{\tau}^\dagger, \mathbf{z}^\dagger)$ whenever the minimum strictly improves.

Lemma 5.2. *For every optimal leader tax $\hat{\tau}^*$ and every $\mathbf{z}^* \in \Psi(\hat{\tau}^*)$, $\mathrm{UB}^{(k)} \geq E(\mathbf{z}^*)$ for every k.*

Proof. Immediate: $\mathrm{UB}^{(k)}$ is the emissions of an explicitly bilevel-feasible solution, and $E(\mathbf{z}^*)$ is optimal among such solutions by Assumption A11. □

5.3 Deterministic bounding procedure

The full DBP is stated as Algorithm 1.

Algorithm 1: Deterministic Bounding Procedure (DBP)

Input: Tax bounds $[\underline{\hat{\tau}}, \overline{\hat{\tau}}]$; tolerance $\epsilon > 0$.

Output: Tax index $\hat{\tau}^\dagger$ and bilevel-feasible response $\mathbf{z}^\dagger$ with Gap $\leq \epsilon$.

// Initialisation. The relaxation master optimises over the same feasible region $\mathcal{X}$ as the follower (Section 4.5), which does *not* depend on $\hat{\tau}$ - only the follower's objective (4.7) does. This is what lets a cut generated at one tax index remain valid at every other (Theorem 5.1).

LB $\leftarrow -\infty$; UB $\leftarrow +\infty$; $\mathcal{C} \leftarrow \emptyset$

$(\hat{\tau}^{(0)}, \cdot) \leftarrow \arg\min_{\hat{\tau} \in \mathcal{T}^\tau, \mathbf{z} \in \mathcal{X}} E(\mathbf{z})$

$k \leftarrow 0$

while UB $-$ LB $> \epsilon \cdot$ UB **do**

  // -- Upper bound: exact follower evaluation at $\hat{\tau}^{(k)}$ --

  // Evaluate the follower exactly at the master's current candidate $\hat{\tau}^{(k)}$ Solve (4.7) at $\hat{\tau} = \hat{\tau}^{(k)}$ to optimality, and among its optimal responses select the one with least emissions (Assumption A11), giving $\mathbf{z}^{(k)} \in \Psi(\hat{\tau}^{(k)})$, $c_k := C(\mathbf{z}^{(k)})$, $e_k := E(\mathbf{z}^{(k)})$

  if $e_k <$ UB **then**

    UB $\leftarrow e_k$; $\hat{\tau}^\dagger \leftarrow \hat{\tau}^{(k)}$; $\mathbf{z}^\dagger \leftarrow \mathbf{z}^{(k)}$

  // Record the cut $C(\mathbf{z}) + \frac{\hat{\tau}}{100} E(\mathbf{z}) \leq c_k + \frac{\hat{\tau}}{100} e_k$, valid for every $\hat{\tau} \in \mathcal{T}^\tau$ since $\mathbf{z}^{(k)} \in \mathcal{X}$ regardless of $\hat{\tau}$ (Theorem 5.1)

  $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\hat{\tau}^{(k)}, c_k, e_k)\}$

  // -- Lower bound: re-solve the relaxation master with the new cut --

  // Re-solve the relaxation master with all cuts collected so far

  $(\hat{\tau}^{(k+1)}, \mathbf{z}) \leftarrow \arg\min_{\hat{\tau} \in \mathcal{T}^\tau, \mathbf{z} \in \mathcal{X}} \Big\{ E(\mathbf{z}) \ : \ C(\mathbf{z}) + \frac{\hat{\tau}}{100} E(\mathbf{z}) \leq c_j + \frac{\hat{\tau}}{100} e_j \ \ \forall (\hat{\tau}^{(j)}, c_j, e_j) \in \mathcal{C} \Big\}$

  LB $\leftarrow E(\mathbf{z})$

  Gap $\leftarrow$ (UB $-$ LB)/UB

  $k \leftarrow k + 1$

return $\hat{\tau}^\dagger$, $\mathbf{z}^\dagger$, Gap

5.4 Convergence analysis

Theorem 5.3 (Convergence of DBP)**.** *Under Assumptions A1, A2, A7 and A11, the DBP terminates in a finite number of iterations and returns a solution with optimality gap* Gap $\leq \epsilon$.

Proof. See Section A. □

Remark 5.4 (Iteration complexity). Each iteration evaluates one tax index exactly. Once cut (5.1) for index $\hat{\tau}^{(j)}$ has been added, re-proposing $\hat{\tau} = \hat{\tau}^{(j)}$ can no longer yield a relaxation value below the optimistic value e_j already found there (Section A). Since $\mathcal{T}^{\tau} = \{\underline{\hat{\tau}}, \ldots, \overline{\hat{\tau}}\}$ is finite, the procedure therefore terminates after at most $\overline{\hat{\tau}} - \underline{\hat{\tau}} + 1$ iterations even at $\epsilon = 0$, with no dependence on a refinement step size and no breakpoint-density argument required.

5.5 Implementation

The DBP is implemented in Python using `gurobipy`; the reproducibility environment uses Python 3.14.4 and Gurobi 13.0.1. The command-line entry point `tools/solver/solve_batch.py` exposes several leader/follower oracles. The bilevel results in Section 6 use the lexicographic exact workflow `lf_lex_exact` — the lexicographic leader of (4.6) over a follower (4.7) that is solved to optimality with ties broken toward lower emissions (Assumption A11). On the base instances the workflow is cross-checked against an independent exhaustive enumeration of the tax index, `enum_lf_lex_exact`, which returns identical leader decisions.

The reported runs use relative DBP tolerance 10^{-4}, per-subproblem Gurobi MIP gap 10^{-4}, six solver threads, and no per-instance time limit; outputs are written under `results/runs/` and every saved solution is revalidated by an independent feasibility-and-objective check before being stored. We therefore report each run as *validated* at a relative gap below 10^{-4}: termination is by the relative-gap criterion of Algorithm 1 rather than by an exactness certificate.

6 Computational Experiments

We use the model and the DBP of Sections 4 and 5 to answer four questions. *What* carbon tax does the regulator choose, and what cost–emission trade-off does it induce (Section 6.2)? *Which* model inputs drive cost and emissions (Section 6.3)? *Which* design levers — cleaner technology or transport — actually abate emissions (Section 6.4)? And does the DBP deliver these answers reliably (Section 6.5)? All reported numbers are regenerated from the saved solver outputs by `tools.experiments.sensitivityanalyze-all` and stored under `results/analysis`; the follower-objective identity $C + \tau E =$ follower objective holds across all runs (maximum residual 6×10^{-8}).

6.1 Experimental setup

Instances. The benchmark is produced by the repository generator. Twelve base codes are defined as the Cartesian product of two network sizes (I17, I25), two material/product scales (M4, M8), and three planning horizons (T3, T6, T12), under the naming convention `STD_I`n`_M`m`_T`t (Table 7). The eight T3 and T6 instances are solved to validation and form the basis of the analysis; the four T12 instances are intractable within our single-machine compute budget and are excluded throughout.

Table 7. Generated benchmark-instance codes.

Code	Values	Implemented meaning
I	17, 25	Total entity nodes in $\mathcal{S}\cup\mathcal{F}\cup\mathcal{W}\cup\mathcal{R}\cup$AIR$\cup$PORT. I17 uses 3 suppliers, 3 factories, 3 warehouses, 4 retailers, 2 airports, and 2 seaports; I25 increases warehouses to 9 and airports to 4.
M	4, 8	Material/product scale. M4 uses 2 raw materials and 1 final product; M8 uses 4 raw materials and 2 final products.
T	3, 6, 12	Number of planning periods in the generated instance.
Modes	mrd1–mrd2 ($\mathcal{M}^{\mathrm{rd}}$), ma1–ma2 ($\mathcal{M}^{\mathrm{a}}$), ms1–ms2 ($\mathcal{M}^{\mathrm{s}}$)	Common transport catalogue with two alternatives per family, following Tautenhain et al. The road modes ($\mathcal{M}^{\mathrm{rd}}$) are company-owned truck alternatives; the air ($\mathcal{M}^{\mathrm{a}}$) and sea ($\mathcal{M}^{\mathrm{s}}$) modes are outsourced carrier alternatives.
$\hat{\tau}$	0,. . . ,20	Discrete tax index. The effective tax is $\tau = 0.01\,\hat{\tau}$ USD/kg CO_2e, so the implemented domain is $[0.00, 0.20]$.

Scenario design. To probe the supply chain's design levers, each T3 base instance is expanded in three ways: a *technology* scenario that enlarges the production and remanufacturing technology menu, a *transport* scenario that enlarges the transport-mode catalogue, and a *combined* scenario that applies both. This yields a 2×2 design (technology on/off $\times$ transport on/off) per instance.

Sensitivity design. A one-at-a-time (OAT) screen perturbs eleven scalar input factors — together with several structural variants (e.g., removal of the mandatory-return floor) — around four base instances, for 75 sensitivity runs in total. Each output change is normalized to an elasticity so that factors are comparable. The screen is complemented by a Morris elementary-effects design [Morris, 1991, Saltelli et al., 2010] on `STD_I17_M4_T3` and by extended tax-range runs that lift the leader ceiling from $\overline{\hat{\tau}} = 20$ to 40 and 80, used in Section 6.2 to test whether the base range truncates the regulator's optimum.

Solver configuration. All runs use the lexicographic exact workflow `lf_lex_exact` described in Section 5.5, with relative DBP tolerance 10^{-4}, per-subproblem Gurobi MIP gap 10^{-4}, six threads, and no time limit; base runs are cross-checked against the independent enumeration `enum_lf_lex_exact`.

6.2 Optimal carbon tax and the cost–emission trade-off

Table 8 reports the eight validated base runs.

Table 8. Validated lexicographic DBP runs on the base instances.

Instance	$\tau^\star$	$E^\star$	$C^\star$	Iter.	Rel. gap	Time (s)
STD_I17_M4_T3	0.04	32,231	632	2	8.68e-09	36.7
STD_I17_M4_T6	0.03	99,044	1,544	2	7.56e-10	51.8
STD_I17_M8_T3	0.20	119,257	635	3	1.05e-06	127.0
STD_I17_M8_T6	0.20	224,362	2,237	2	8.31e-10	1,936.0
STD_I25_M4_T3	0.15	15,898	431	3	8.85e-12	37.6
STD_I25_M4_T6	0.02	31,533	910	2	1.61e-09	138.0
STD_I25_M8_T3	0.02	130,002	982	2	8.02e-16	152.6
STD_I25_M8_T6	0.07	144,754	2,118	2	7.17e-10	2,367.0

Notes: the eight solved base instances, method `lf_lex_exact`, under `results/runs/base/*/lf_lex_exact`. All rows have `success=True`, are `validated`, and terminate on the relative-gap criterion with gap $\leq 10^{-4}$. $\tau^\star$ is in USD/kg CO_2e; $E^\star$ in 10^3 kg CO_2e and $C^\star$ in 10^3 monetary units. The two M8 T3/T6 rows with $\tau^\star = 0.20$ sit at the base tax ceiling $\bar{\tau} = 20$ and are re-examined under extended ranges in Section 6.2.

The regulator's optimal tax is strongly *instance-specific*: across the eight networks $\tau^\star$ spans the full base range, from 0.02 to 0.20 USD/kg CO_2e. There is no single "right" carbon price; the rate that minimises emissions for one network configuration can be an order of magnitude away from the rate for another. Because the follower carries binary facility, technology, capacity, and transport decisions, its response shifts in discrete blocks rather than smoothly, so nearby tax levels can induce the same network and identical emissions.

This blockiness has a sharp consequence at the high end. Two instances (`STD_I17_M8_T3` and `STD_I17_M8_T6`) place $\tau^\star$ exactly at the base ceiling $\bar{\tau} = 20$, signalling that the base range may truncate the true optimum. The extended-range runs (Section B.1) reveal a *saturating* emission–tax response: beyond a moderate rate, additional tax buys negligible further abatement. For `STD_I25_M8_T3`, lifting the ceiling moves the lexicographic optimum from $\hat{\tau} = 2$ to $\hat{\tau} = 36$ while cutting emissions by only 0.04% — an eighteen-fold higher tax for a rounding-error reduction. For `STD_I25_M4_T3` the optimum moves from $\hat{\tau} = 15$ to 31 for a 0.016% reduction, and for `STD_I17_M8_T3` emissions are flat to within solver tolerance across $\hat{\tau} \in [14, 31]$, so the emissions-minimising tax is not point-identified at all but spread over a wide band.

This is the central caution of the study. A regulator whose *sole* objective is emissions minimisation is ill-posed on a saturating frontier: it can rationalise an arbitrarily heavy tax for vanishing environmental gain. The lexicographic structure of (4.6) — minimise emissions, then levy the lowest rate that attains the minimum — is what disciplines this pathology, selecting the bottom of the near-flat band; but it does so only up to the solver's relative-gap tolerance. The policy reading is that the emissions target should be paired with an explicit cost criterion (or a minimum meaningful-abatement threshold) rather than pursued in isolation.

6.3 What drives cost and emissions

Table 9 ranks the input factors by their normalized influence on cost and on emissions.

The two outputs are governed by different factors. *Cost* responds most to production

Table 9. Normalized local sensitivity (OAT elasticities).

Input factor	Cost elast.	Emissions elast.
Production capacity	0.58	0.04
Demand	0.57	0.98
Production emission intensity	0.12	0.56
Return availability	0.09	<0.01
Return processing cost	0.05	<0.01
Scope-3 (supply) emissions	0.04	0.42
Bill-of-materials ratio	0.03	<0.01
Warehouse cost, truck capacity, tax-range ceiling, air/sea cost	≤0.02	<0.01

Notes: mean absolute normalized elasticities across the four screened base instances (I17/I25 × M4/M8, $T = 3$); a value of 1.0 means a 1% input change moves the output by 1%. Structural perturbations without a scalar input change carry no elasticity; 31 of the screened perturbations fall below the 0.05% solver-noise floor.

capacity (elasticity 0.58) and demand (0.57); *emissions* respond almost one-for-one to demand (0.98) and strongly to the production and scope-3 (upstream supply) emission intensities (0.56 and 0.42). The ranking is stable across network sizes — demand is the top cost factor in three of the four screened instances (production emission intensity in the fourth) — which strengthens the generality of the finding. The Morris screen agrees that demand dominates, though as a single-trajectory screen on one instance it is preliminary and we treat it as confirmatory only.

The economically important message is what does *not* matter: transport cost, truck capacity, warehouse cost, and the air/sea cost all carry emissions elasticities below 0.01. Emissions in these networks are determined by *what is produced and procured*, not by how goods are moved. This directly motivates the design-lever experiment that follows.

6.4 Design levers: technology versus transport

Table 10 contrasts the three design-lever scenarios against the base instances.

Table 10. Optimal-response emissions under the design-lever scenarios.

Instance	Base $E^\star$	Technology	Transport	Combined
STD_I17_M4_T3	32,231	22,083 (−31.5%)	32,242 (+0.03%)	22,092 (−31.5%)
STD_I17_M8_T3	119,257	80,440 (−32.5%)	119,298 (+0.03%)	80,515 (−32.5%)
STD_I25_M4_T3	15,898	11,049 (−30.5%)	15,920 (+0.14%)	11,054 (−30.5%)
STD_I25_M8_T3	130,002	96,661 (−25.6%)	130,023 (+0.02%)	96,725 (−25.6%)

Notes: optimal-response emissions $E^\star$ (10^3 kg CO_2e) under the base instance and three design-lever scenarios; parentheses give the change relative to the base. *Technology* expands the production/remanufacturing technology menu, *Transport* expands the transport-mode catalogue, and *Combined* applies both. Expanding technology drives essentially all of the abatement; expanding transport modes alone is inert.

Expanding the production and remanufacturing technology menu reduces optimal emissions by 26% to 33% across all four instances. Expanding the transport-mode catalogue alone changes emissions by essentially zero (at most +0.1%), and the combined scenario is indistinguishable

from technology alone. Cleaner production and remanufacturing capability, not modal choice, is the binding abatement lever in these networks — exactly the pattern the elasticities predicted. The cost side is more nuanced: the technology menu can raise cost slightly on the smaller I17 instances (+2 to +4%) but sharply *reduces* both cost and emissions on the larger I25 networks (e.g., `STD_I25_M4_T3`, where the combined scenario cuts cost by 42% while abating 30.5% of emissions), indicating that for larger networks the expanded technology set unlocks configurations that are jointly cheaper and cleaner.

6.5 Solver behaviour and validation

The findings above rest on the DBP returning trustworthy solutions across the entire experimental space. It does: all 95 committed runs are validated, the largest relative gap is 1.9×10^{-5} (and 1.0×10^{-6} on the base instances), none exceeds the target tolerance, and the procedure converges in two to three iterations on average. Table 11 profiles runtime by planning horizon.

Table 11. Runtime profile of the validated base runs.

Planning horizon	Solved runs	Min (s)	Median (s)	Max (s)
$T = 3$	4	36.7	82.3	152.6
$T = 6$	4	51.8	1,037.0	2,367.0

Notes: wall-clock time of the eight validated base runs in Table 8, grouped by planning horizon. The four $T = 12$ base instances are intractable within the compute budget and are not solved. Runtime rises steeply with the horizon and with the larger I25–M8 network.

Runtime is driven by the planning horizon and the larger I25–M8 network: T3 instances solve in under three minutes, T6 instances in up to forty minutes, and the most demanding run in the whole study — the `STD_I25_M8_T3` combined scenario — takes 1.86×10^4 s (≈ 5.2 h). The four T12 base instances do not solve within the compute budget, which bounds the present study to the T3–T6 range and marks horizon length as the primary computational bottleneck.

7 Discussion

The experiments in Section 6 support three substantive conclusions about taxing closed-loop supply chains: the emissions-minimising tax is instance-specific and sits on a saturating frontier; cleaner production and remanufacturing technology, not transportation, is the binding abatement lever; and emissions are governed by demand and by production and upstream emission intensities. We discuss what each implies for managers and regulators, then state the limits of the analysis.

7.1 The cost–emission trade-off and the saturating frontier

The optimal taxes recovered by the bilevel model vary by an order of magnitude across network configurations ($\tau^\star \in [0.02, 0.20]$), and the follower's binary facility, technology, capacity, and transport decisions make its response change in discrete blocks rather than smoothly. The

most consequential feature is the *saturation* of the emission–tax response: beyond a moderate rate, further taxation buys almost no additional abatement, so the emissions-minimising tax is attained over a wide, near-flat band (Section 6.2). A regulator that optimises emissions alone is therefore ill-posed in this regime — the objective alone cannot distinguish a moderate tax from a punitive one that achieves a rounding-error reduction. The lexicographic least-rate criterion is what restores well-posedness, and the practical lesson is that an emissions target must be paired with a cost criterion or a minimum meaningful-abatement threshold; otherwise the "optimal" tax is an artifact of where the grid happens to end.

7.2 Robustness of the qualitative findings

Two features make the findings more than instance anecdotes. The factor rankings are *stable across network sizes* — demand is the leading cost driver in three of four screened instances, and technology dominates abatement in all four scenario instances — so the qualitative conclusions are not artifacts of a single configuration. And the underlying solves are trustworthy: every one of the 95 committed runs is validated, the largest relative gap is 1.9×10^{-5}, and the procedure converges in two to three iterations. The robust claim is thus not that one tax rate is universal, but that *which lever abates and which inputs matter* is consistent across the tested networks.

7.3 Managerial implications

For supply-chain managers, the clearest signal is where to invest abatement effort. Expanding the production and remanufacturing technology menu reduces optimal emissions by 26–33%, while expanding the transport-mode catalogue is effectively inert; on the larger I25 networks the richer technology set is in fact jointly cheaper and cleaner. Capital and managerial attention are therefore better directed at cleaner production and remanufacturing capability than at modal shift in these networks. The sensitivity analysis sharpens the point: demand and the production and upstream (scope-3) emission intensities — *what is made and procured* — dominate both cost and emissions, so demand exposure and supplier emission intensity are the key risks to manage, whereas trucking and storage parameters are second-order.

7.4 Policy implications

For regulators, the instance-specific variation in $\tau^\star$ argues against a one-size-fits-all carbon price: if sectoral network structure is known, the bilevel model supports differentiated calibration, and if it is not, it still provides a reproducible way to test representative networks before setting a rate. The saturating frontier carries a second, cautionary message: once a tax is high enough to trigger the available abatement, raising it further imposes cost on the supply chain for negligible environmental return. A tax instrument is most effective when calibrated to the supply chain's abatement *response*, which is exactly what the bilevel formulation makes observable; pushing past that point is economically wasteful. Because the binding abatement lever here is production/remanufacturing technology, a carbon tax may also be more effective

when paired with instruments that lower the cost of adopting cleaner technology.

7.5 Limitations of the present study

Several modelling and computational choices bound the analysis.

Generated deterministic data. Demand, return fractions, costs, emissions, and capacities are generated and deterministic. The results are reproducible but are not calibrated to a specific industrial case, so the magnitudes (e.g., the 26–33% technology effect) should be read as model-internal rather than as field estimates.

Lexicographic and optimistic tie-breaking. The leader minimises emissions and then the tax, and the follower breaks cost ties toward lower emissions. These optimistic conventions favour lower-emission outcomes; a pessimistic regulator that cannot assume favourable follower behaviour would face a less forgiving problem, which we do not model here.

Bounded benchmark and horizon. The study covers the eight T3–T6 base instances; the four T12 instances are intractable within the compute budget, so the longest horizons are out of reach without further algorithmic work. The sensitivity screen covers four bases and the Morris design covers one, so the global screening is preliminary.

Structural observations. Warehouses never open in any optimal response under the generated data, so the warehouse echelon is effectively dormant in these instances — a property of the data and cost structure rather than evidence that storage is never useful. We also do not decompose abatement into per-facility or per-technology contributions; the scenario design isolates the technology *lever* but not the individual mechanisms within it.

Single instrument and no external benchmark. The leader wields one discrete tax index over $\{0, \ldots, \overline{\hat{\tau}}\}$; caps, subsidies, cap-and-trade, and dynamic trajectories are outside the present scope. We also do not benchmark the DBP against an alternative exact method (e.g., a KKT reformulation or exhaustive grid beyond the enumeration cross-check), so we make no cross-method performance claim.

8 Conclusion

This paper has addressed the design of sustainable closed-loop supply chains under a carbon-tax policy by modelling the interaction between the regulator and the supply-chain operator as a bilevel mixed-integer programme. The regulator lexicographically minimises emissions and then the tax rate, while the operator solves a multi-echelon, multi-period CLSC design and operations problem with integer facility, technology, capacity, and transportation decisions. We developed a deterministic bounding procedure (DBP) that combines a relaxation-based lower bound with exact follower evaluations and returns validated bilevel solutions on the generated instances.

8.1 Summary of findings

Three findings emerge from the computational study:

1. **The emissions-minimising tax is instance-specific and sits on a saturating frontier.** Optimal rates span 0.02–0.20 USD/kg CO_2e, and beyond a moderate level extra tax yields negligible abatement, so a purely emissions-driven regulator is ill-posed unless paired with a cost or meaningful-abatement criterion.
2. **Technology, not transport, is the binding abatement lever.** Expanding the production and remanufacturing technology menu cuts optimal emissions by 26–33%, while expanding the transport-mode catalogue is essentially inert; emissions are driven by demand and by production and upstream (scope-3) emission intensities.
3. **The DBP is reliable but horizon-bound.** All 95 committed runs are validated at relative gaps no larger than 1.9×10^{-5} in two to three iterations; T3–T6 instances solve within hours, whereas the T12 instances are intractable within the compute budget, marking horizon length as the main computational bottleneck.

8.2 Academic contributions

The contributions of the paper are threefold. First, it formulates a bilevel CLSC model that integrates a discrete, lexicographic regulatory tax decision with a detailed integer supply-chain design problem. Second, it develops a convergent DBP with saved-solution validation and uses it to derive policy- and management-relevant insight — the saturating emission–tax frontier and the primacy of the technology lever — that single-level models cannot reveal. Third, it provides a tested generated benchmark, a sensitivity and scenario suite, and a documented result schema that ties every reported number to saved solver outputs.

8.3 Limitations and future research directions

Several extensions deserve attention.

Cost-aware leader objectives. The saturating frontier makes a purely emissions-minimising leader ill-posed; a natural next step is a leader objective that trades emissions against tax burden explicitly, or that imposes a minimum meaningful-abatement threshold, so the optimal tax is well-identified without relying on the lexicographic tie-break.

Scalability to longer horizons. The T12 instances are out of reach within the present compute budget; decomposition, warm starts, and parallel DBP implementations are the most promising routes to the longest horizons and larger networks.

External benchmark implementations. KKT reformulations, exhaustive bilevel grid search, or other benchmark methods should be implemented before any cross-method performance claims are made.

Mechanism-level diagnostics. The scenario design isolates the technology lever but not the individual mechanisms within it; decomposing abatement by facility, technology, remanufacturing, inventory, and transport decision would support finer mechanism-specific claims.

Stochastic and robust extensions. Embedding demand, return-rate, cost, and emission uncertainty into the follower problem would broaden the model's applicability to volatile markets.

Pessimistic and risk-averse leader models. Implementing pessimistic tie-breaking and risk-averse leader objectives would strengthen the regulatory interpretation when the regulator cannot assume favourable follower tie-breaking.

Mixed regulatory instruments. Caps, subsidies, cap-and-trade, and dynamic tax trajectories are natural extensions of the leader programme.

Algorithmic enhancements. Adaptive sampling, decomposition, warm starts, and parallel DBP implementations are promising routes for larger instances.

Pursuing these extensions would turn the current reproducible prototype into a broader decision-support framework for joint supply chain and climate-policy design.

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Author Contributions

Marcelo Pinto: Conceptualisation, Methodology, Software, Formal Analysis, Writing – Original Draft. **Paulo Nascimento**: Methodology, Validation, Writing – Review & Editing. **Cristóvão Silva**: Supervision, Writing – Review & Editing. **Maria João Alves**: Methodology, Formal Analysis, Writing – Review & Editing. **Carlos Henggeler**: Supervision, Writing – Review & Editing. **Samuel Moniz**: Conceptualisation, Supervision, Project Administration, Funding Acquisition, Writing – Review & Editing.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A Proofs

Proof of Theorem 5.1. Fix $k \geq 0$ and let $(\hat{\tau}^*, \mathbf{z}^*)$ be optimal for (4.6), so $\mathbf{z}^* \in \Psi(\hat{\tau}^*)$. The feasible region $\mathcal{X}$ of the follower's problem does not depend on $\hat{\tau}$ (Section 4.5); only the objective (4.7) does. Hence every $\mathbf{z}^{(j)} \in \Psi(\hat{\tau}^{(j)})$ used to generate cut j, for j ranging over the cut set $\mathcal{C}$ accumulated by iteration k, is itself a feasible – if possibly suboptimal – point of the follower's problem at $\hat{\tau}^*$. Optimality of $\mathbf{z}^*$ at $\hat{\tau}^*$ therefore gives

$$C(\mathbf{z}^*) + \tfrac{\hat{\tau}^*}{100} E(\mathbf{z}^*) \;\leq\; C(\mathbf{z}^{(j)}) + \tfrac{\hat{\tau}^*}{100} E(\mathbf{z}^{(j)}) = c_j + \tfrac{\hat{\tau}^*}{100} e_j \qquad \text{for every } (\hat{\tau}^{(j)}, c_j, e_j) \in \mathcal{C}.$$

Thus $(\hat{\tau}^*, \mathbf{z}^*)$ satisfies cut (5.1) for every cut accumulated by iteration k, i.e. it is feasible for the relaxation master defining $\mathrm{LB}^{(k)}$ in (5.2). Since $\mathrm{LB}^{(k)}$ minimises $E(\cdot)$ over that (larger) feasible set, $\mathrm{LB}^{(k)} \leq E(\mathbf{z}^*)$. □

Proof of Theorem 5.3. **Validity of bounds.** By Theorems 5.1 and 5.2, the inequalities $\mathrm{LB}^{(k)} \leq E(\mathbf{z}^*) \leq \mathrm{UB}^{(k)}$ hold for every iteration k of Algorithm 1.

Finite termination. Fix any tax index $\hat{\tau}' \in \mathcal{T}^\tau$ already evaluated by iteration k, i.e. $\hat{\tau}' = \hat{\tau}^{(j)}$ for some cut $(\hat{\tau}^{(j)}, c_j, e_j) \in \mathcal{C}$. Evaluating cut (5.1) at $\hat{\tau} = \hat{\tau}'$ gives $C(\mathbf{z}) + \frac{\hat{\tau}'}{100} E(\mathbf{z}) \leq c_j + \frac{\hat{\tau}'}{100} e_j$ for every $\mathbf{z}$ feasible in the relaxation master defining $\mathrm{LB}^{(k)}$; and since $c_j + \frac{\hat{\tau}'}{100} e_j = C(\mathbf{z}^{(j)}) + \frac{\hat{\tau}'}{100} E(\mathbf{z}^{(j)})$ is itself the optimal value of (4.7) at $\hat{\tau}'$ (because $\mathbf{z}^{(j)} \in \Psi(\hat{\tau}')$), no feasible $\mathbf{z}$ can improve on $\mathbf{z}^{(j)}$ in objective (4.7) at $\hat{\tau}'$. Hence any $\mathbf{z}$ chosen by the master at $\hat{\tau} = \hat{\tau}'$ must itself lie in $\Psi(\hat{\tau}')$, and minimising $E(\cdot)$ over that restriction recovers exactly $\min\{E(\mathbf{z}) : \mathbf{z} \in \Psi(\hat{\tau}')\}$, the optimistic value at $\hat{\tau}'$ (Assumption A11) – i.e. the master's bound at every previously-evaluated index is *exact*, not merely a relaxation. Since $\mathcal{T}^\tau = \{\underline{\hat{\tau}}, \ldots, \overline{\hat{\tau}}\}$ has only $\overline{\hat{\tau}} - \underline{\hat{\tau}} + 1$ elements, after at most $\overline{\hat{\tau}} - \underline{\hat{\tau}} + 1$ iterations every admissible tax index either (i) has been evaluated exactly, making $\mathrm{LB}^{(k)}$ exact there, or (ii) is dominated, in the master's emissions-minimisation, by an index covered by (i). In either case $\mathrm{LB}^{(k)} = \mathrm{UB}^{(k)}$ once $k = \overline{\hat{\tau}} - \underline{\hat{\tau}} + 1$, so the gap closes within a finite, explicitly bounded number of iterations – with no continuity or breakpoint-density argument on $E(\mathbf{z}^*(\hat{\tau}))$ required.

ϵ-optimality. On termination, $\mathrm{Gap} = (\mathrm{UB}^{(k)} - \mathrm{LB}^{(k)})/\mathrm{UB}^{(k)} \leq \epsilon$, so the returned index $\hat{\tau}^\dagger$ satisfies $E(\mathbf{z}^\dagger) = \mathrm{UB}^{(k)} \leq \mathrm{LB}^{(k)}/(1-\epsilon) \leq E(\mathbf{z}^*)/(1-\epsilon)$, establishing the claimed ϵ-optimality guarantee. □

B Additional Computational Results

B.1 Extended tax-range analysis

Section 6.2 reports that two base instances place the optimal tax at the base ceiling $\overline{\hat{\tau}} = 20$ and that the emission–tax response saturates. Table 12 substantiates this by re-solving the four screened base instances with the leader ceiling lifted to 40 and 80. Lifting the ceiling moves the lexicographic optimum to a much higher tax index on three of the four instances, yet the corresponding emissions barely change: the largest abatement unlocked by quadrupling the admissible tax is 0.07%. For `STD_I17_M8_T3` the emissions-minimising index is not even

point-identified — it shifts from 14 to 31 between the two extended ranges while emissions stay flat to within solver tolerance.

Table 12. Lexicographic optimum under extended leader tax ceilings.

Instance	Base $\hat{\tau}\,(E^\star)$	$\overline{\hat{\tau}}$=40 $\hat{\tau}\,(E)$	$\overline{\hat{\tau}}$=80 $\hat{\tau}\,(E)$	Extra abate.
STD_I17_M4_T3	4 (32,231)	4 (32,231)	61 (32,209)	−0.07%
STD_I17_M8_T3	20 (119,257)	14 (119,258)	31 (119,257)	<0.01%
STD_I25_M4_T3	15 (15,898)	31 (15,896)	31 (15,896)	−0.02%
STD_I25_M8_T3	2 (130,002)	36 (129,949)	36 (129,949)	−0.04%

Notes: $\hat{\tau}$ is the optimal integer tax index and E the induced emissions (10^3 kg CO_2e); the last column is the emissions reduction relative to the base-ceiling optimum. Source: `results/runs/.../oat/<base>/tax_range_max040` and `tax_range_max080`.

B.2 Solver coverage and external benchmarks

The numerical study rests on 95 validated runs: the eight base instances and twelve design-lever scenarios are each solved with the lexicographic exact workflow `lf_lex_exact` and, on the base instances, cross-checked against the independent enumeration `enum_lf_lex_exact` (identical leader decisions); the remaining 75 runs form the sensitivity suite. No result directory or solver entry point implements a KKT single-level reformulation or an exhaustive-grid bilevel benchmark beyond this enumeration cross-check, so we make no cross-method performance comparison.

C Instance Data and Parameter Values

The committed benchmark data are the JSON files under `data/instances/base`. The generator entrypoint is `data/generator/instance.py` (`generate_instance`/ `write_instance`), invoked via `python -m data.generator benchmark`; sampling distributions and bounds are implemented in `data/generator/tautenhain_data.py`, with the instance-tree layout documented in `data/README.md`.

C.1 Benchmark Scope

The current base benchmark contains 12 instances:

$$\{\mathrm{I17}, \mathrm{I25}\} \times \{\mathrm{M4}, \mathrm{M8}\} \times \{\mathrm{T3}, \mathrm{T6}, \mathrm{T12}\}.$$

The `STD` prefix is the only prefix generated by the current benchmark script. The eight T3 and T6 instances are solved to validation; the four T12 instances are generated but not solved, being intractable within the compute budget (Section 6.5).

Table 13. Implemented benchmark-generation settings.

Setting	Implemented value
Suppliers	3
Factories	3
Retailers	4
Warehouses	3 for I17; 9 for I25
Airports	2 for I17; 4 for I25
Seaports	2
Raw materials and products	M4: 2 raw materials and 1 product; M8: 4 raw materials and 2 products
Production technologies	3
Remanufacturing technologies	3
Planning periods	T3, T6, or T12
Coordinate span	750 km square
Tax-index domain	$\hat{\tau} \in \{0, \ldots, 20\}$
Effective tax rate	$\tau = 0.01\,\hat{\tau}$ USD/kg CO_2e

C.2 Seed Protocol

Benchmark seeds are deterministic functions of the benchmark prefix and suffix. The implementation hashes `seed_base:prefix:suffix` with SHA-256 and uses the first eight hexadecimal digits as the integer seed. The generated seed is stored in each instance under `metadata.benchmark.seed`.

C.3 Data Availability

All instance files, generation scripts, solver scripts, tests, and saved result artifacts are part of the accompanying repository. The reported numerical results should be regenerated with:

```
python -m pytest -q
python tools/solver/solve_batch.py data/instances/base/STD_I17_M4_T3.json
python -m tools.experiments.sensitivity analyze-all
```

The final command regenerates every derived table, figure, and report under `results/analysis` from the saved solver outputs, without re-solving.